

Chapter-6

5. Elementary Probability

Introduction

A proportion is defined to represent the relative size of the portion of a population with a certain (binary) characteristic. For example, disease prevalence is the proportion of a population with a disease. Similarly, we can talk about the proportion of positive reactors to a certain screening test, the proportion of males in colleges, and so on. A proportion is used as a descriptive measure for a target population with respect to a binary or dichotomous characteristic. It is a number between 0 and 1 (or 100%); the larger the number, the larger the subpopulation with the characteristics [e.g., 70% male means more males (than 50%)].

Even science is uncertain. Scientists are sometimes wrong. They arrive at different conclusions in many different areas: the effects of a certain food ingredient or of low-level radioactivity, the role of fats in diets, and so on. Many studies are inconclusive. For example, for decades surgeons believed that a radical mastectomy was the only treatment for breast cancer. More recently, carefully designed clinical trials showed that less drastic treatments seem equally effective. Why it is that science is not always certain? Nature is complex and full of unexplained biological variability. In addition, almost all methods of observation and experiment are imperfect. Observers are subject to human bias and error. Science is a continuing story; subjects vary; measurements fluctuate.

Basic terms in Probability

Experiment: - It is an activity or a trial that leads to well-defined results called outcomes, but it is uncertain to which outcome will occur. Hence, a probability experiment is identified by two properties. First each experiment has several possible outcomes (two or more) and all these outcomes are known in advance and, second none of these outcomes can be predicted with certainty. For example for the experiment of tossing a fair coin, we cannot be certain whether the outcome will be a head.

A Random experiment is an experiment whose outcome is not known until it is observed. Or it is the process of observing the outcome of a chance event.

An outcome is particular result of an experiment. Or an outcome is the all possible results of the random experiment.

Sample space: is the set of all possible outcomes for the experiment. *Sample space* is simply sets whose elements describe the outcomes of the experiment in which we are interested. Each possible outcome is called sample point. It is denoted by S.

- A sample space may be finite, countable infinite or uncountable.
- A finite or countable infinite sample space is called a discrete sample space.
- An uncountable sample space is called a continuous sample space

Examples:

$S = \{0, 1, 2, 3\}$ (discrete and finite)

$S = \{0, 1, 2, 3, \dots\}$ (discrete, infinite)

$S = \{4.5, 4.6, 4.7\}$ (discrete, finite)

$S = \{HH, HT, TH, TT\}$ (discrete, finite)

$S = [0, 1] = \{\text{all numbers between 0 and 1 inclusive}\}$ (continuous, infinite)

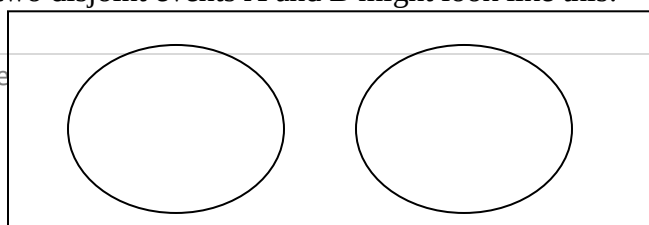
Event: - is an outcome or a set of outcomes (having common characteristics) of an experiment. It is subsets of the sample space. For example getting one head in a trial of tossing three coins simultaneously would be an event, $E = \{HT, TH\}$. Getting even number in rolling a die, $E = \{2, 4, 6\}$.

Example

<u>Experiment</u>	<u>Sample Space</u>
1. Tossing a coin	$S = \{\text{Head (H), Tail (T)}\}$
2. Tossing two coins	$S = \{HH, HT, TH, TT\}$
3. Rolling a die	$S = \{1, 2, 3, 4, 5, 6\}$
4. Selecting an item from a production lot	$S = \{\text{Defective, Non-defective}\}$
5. Introducing a new product	$S = \{\text{Success, Failure}\}$

Mutually exclusive events/Disjoint events: Two events E_1 and E_2 are said to be mutually exclusive events if there is no sample point which is common to both events E_1 and E_2 . That means, $E_1 \cap E_2 = \emptyset$.

Mutually exclusive events are events, which cannot happen at the same time. The Venn diagram to show two disjoint events A and B might look like this:



Ω

A

B

Where Ω -- is a sample space

A --is event A

B—is event B

Note: A sample space, Ω , is a set of outcomes of a random experiment. Every possible outcome must be listed once and only once in the element. For example, if the sample space is $\Omega = \{s_1, s_2, s_3\}$, then each s_i is a sample point.

Definition: Two events E_1 and E_2 are said to be independent if the occurrence of E_1 has no effect on the occurrence of E_2 . That means the knowledge of event E_1 has occurred given no information about the occurrence of the event E_2 . If two events are not independent, they are said to be dependent.

Examples: Experiment: Toss a coin twice and observe the result. Sample space: $= \{HH, HT, TH, TT\}$. An example of a sample point is: HT

Experiment: Toss a coin twice and count the number of heads.

Sample space: $= \{0, 1, 2\}$

Experiment: Toss a coin twice and observe whether the two tosses are the same (E.g. HH or TT).

Sample space: $= \{\text{same, different}\}$

Definition: Event A occurs if we observe an outcome that is a member of the set A.

Note: Ω is a subset of itself, so Ω is an event. The null event empty set, $\emptyset$, is also a subset of Ω . This is called the null event, or the event with no outcomes.

Example: Experiment: throw 2 dice.

Sample space: $\Omega = \{(1, 1), (1, 2), \dots (1, 6), (2, 1), (2, 2), \dots (2, 6), \dots (6, 6)\}$

Event A = “sum of two faces is 5” $= \{(1, 4), (2, 3), (3, 2), (4, 1)\}$

Equally Likely Events: Events which have the same chance of occurring.

Counting rules: Addition, Multiplication, Permutation & Combination rule

In order to calculate probabilities, we have to know

- The number of elements of an event.
- The number of elements of the sample space.

That is in order to judge what is *probable*, we have to know what is *possible*.

In order to determine the number of outcomes one can use several rules of counting:

1. The addition rule
2. The multiplication rule
3. Permutation rule
4. Combination rule

1. The addition Rule

Suppose that a procedure, designated by 1, can be done in n_1 ways. Assume that a second procedure designated by 2, can be done in n_2 ways. Suppose furthermore, that it is not possible that both 1 and 2 done together. Then, the number of ways in which we can do procedure 1 or 2 is $n_1 + n_2$ ways.

Example: suppose we are planning a trip to some place. If there are 3 bus routes & two trains' routes that we can take, then there are $3+2=5$ different routes that we can take.

2. Multiplication rule

If an operation consists of k steps and the 1st step can be done in n_1 ways, the 2nd step can be done in n_2 ways (regardless of how the 1st step was performed), the k^{th} step can be done in n_k ways, (regardless of how the preceding steps were performed), then the entire operation can be performed in $n_1 \cdot n_2 \cdot \dots \cdot n_k$ ways.

Example: Suppose that a person has 2 different pairs of trousers and 3 shirts. In how many ways can he wear his trousers and shirts?

Solution: He can choose the trousers in $n_1 = 2$ ways, and shirts in $n_2 = 3$ ways. Therefore, he can wear in $n_1 \times n_2 = 2 \times 3 = 6$ possible ways.

3. Permutation

An arrangement of objects with attention given to order of arrangement is called permutation. The number of permutation of n different objects taken r at a time is obtained by:

$${}_nP_r = \frac{n!}{(n-r)!} \text{ for } r = 0, 1, 2, \dots, n$$

Permutation Rule

a) The number of permutations of n objects taken all together is $n!$

$$\text{i.e. } n! = n*(n-1)*(n-2)*\dots*3*2*1 = {}_nP_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!$$

Note: By definition $0! = 1$

b) The arrangement of n distinct objects in a specific order using r objects at a time is called the permutation of n objects taken r objects at a time. It is written as ${}_nP_r$ and the formula is

$${}_nP_r = \frac{n!}{(n-r)!}$$

c) The number of distinct permutation of n objects in which n_1 are alike, n_2 are alike... n_k are alike is

$$\frac{n!}{n_1!n_2!\dots n_k!} \text{ where } n = n_1 + n_2 + \dots + n_k$$

Example: Find number of permutations of the letters in the word “**statistics**”.

Solution:

There are 3s's, 3t's, 1a's, 2i's and 1c's. i.e. $n_1 = 3, n_2 = 3, n_3 = 1, n_4 = 2$ and $n_5 = 1$

$$\text{Therefore } \frac{10!}{3!.3!.1!.2!1!} = 50,400.$$

Example: A photographer wants to arrange 3 persons in a row for photograph. How many different types of photographs are possible?

Solution:

Assume 3 persons Aster (A), Lemma (L), Yared (Y) and $n=3$

Since $n! = 3! = 3*2! = 6$, there are 6 possible arrangement ALY, AYL, LAY, LYA, YLA and YAL.

Example: Suppose we have a letters A, B, C, D & E

a) How many permutations are there taking all the four?

b) How many permutations are there taking two letters at a time?

Solution:

a) Here $n = 5$, there are four distinct object.

There are $5! = 120$ permutations.

b) Here $n = 5, r = 2$

There are ${}_5P_2 = \frac{5!}{(5-2)!} = \frac{120}{6} = 20$ permutation

Example: Fifteen Ethiopian athletes were entered to the race. In how many different ways could prizes for the first, the second and the third place be awarded?

Solution

15 Objects taken 3 at a time ${}_{15}P_3 = 15! / (15-3)! = 2730$ ways.

4. Combination

A selection of objects considered without regard to order in which they occur is called Combination. The number of combination of n different objects taking r of them at a time is

$${}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}, \text{ for } r = 0, 1, 2, \dots, n.$$

Example: Given the letters A, B, C, and D list the permutation and combination for selecting two letters.

Solution:

Permutation

AB BA CA DA
AC BC CB DB
AD BD CD DC

Combination

AB BC
AC BD
AD DC

Note that in permutation AB is different from BA but in combination AB is the same as BA.

Example: In a club containing 7 members a committee of 3 people is to be formed. In how many ways can the committee be formed?

Solution: ${}_7C_3 = {}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} \rightarrow {}_7C_3 = \binom{7}{3} = \frac{7!}{3!(7-3)!} = 35$

Example: How many four-digit numbers can be formed with the 10 digits 0,1,2, . . . ,9 if

a/ repetitions are allowed

b/ repetitions are allowed, and

c/ the last digit must be zero & repetitions are not allowed.

Solution:

a/ the first digit can be any one of 9 (since 0 is not allowed). The second, third and fourth digits can be any one of 10. Then $9 \cdot 10 \cdot 10 \cdot 10 = 9000$ numbers can be formed.

b/ the first digit can be any one of 9 & the remaining three can be chosen in ${}_9P_3$ ways. Thus $9 \times {}_9P_3 = 4536$ numbers can be formed.

c/ the first digit can be chosen in 9 ways & the next two digits in ${}_9P_2$ ways. Thus $9 \times {}_9P_2 = 504$ numbers can be formed.

Definition of Probability

Definition: Probability is a numerical measure of the chance or likelihood that a particular event will occur & it lies in the range from 0-1, inclusive. Probability is a building block of inferential statistics.

Definition: Let E be an experiment. Let S be a sample space associated with E. With each event A in S we associate a real number designated by P (A) and called the probability of A.

Generally probability can be divided into two

- i) **Subjective probability:** - probability determined based on individual's own judgment, experience, information, belief . . . is called Subjective probability.
- ii) **Objective probability:** - the probability of an event in a certain experiment based on experimental evidence.

Basic approaches of measuring probability

There are three different conceptual approaches to the study of probability theory.

These are:

- The classical approach
- The frequentist approach.
- The axiomatic approach.

5.1.1. Classical approach

Definition: If there are n equally likely outcomes of an experiment, and out of the n outcomes event A occur only k times the probability of the event A is denoted by P (A) is defined as

$$P(A) = \frac{\text{Number of outcomes favorable to event } A}{\text{Total number of outcomes}} = \frac{n(A)}{n(S)} = \frac{k}{n}$$

Note: Classical approach of measuring probability fails to answer for the following conditions:

- If total number of outcomes is infinite or if it is not possible to enumerate all elements of the sample space.

- If each outcome is not equally likely.

Example: Compute a/ the probability of having two boys & one girl in a three child family using the classical method, assuming boys & girls are equally likely.

b/ using (a) compute the probability of having three boys in a three-child family.

c/ using (a) compute the probability of having three girls in a three –child family.

d/ using (a) compute the probability of having two girls & one boy in three child family.

Solution

The sample space S or the experiment is

$$S = \{BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG\}$$

$$\text{So } n(S) = 8$$

a/ For the event $A = \text{"two boys \& a girl"} = \{BBG, BGB, GBB\}$, we have $n(A) = 3$, Since the outcome are equally likely, the probability of A is $P(A) = n(A)/n(S) = 3/8 = 0.375$

b/ Compute the probability of having three boys in a three-child family.

For the event $B = \text{"three boys"} = \{BBB\}$, we have $n(B) = 1$, Since the outcome are equally likely, the probability of B is $P(B) = n(B)/n(S) = 1/8 = 0.125$

c/ compute the probability of having three girls in a three –child family.

For the event $C = \text{"three girls"} = \{GGG\}$, we have $n(C) = 1$, Since the outcome are equally likely, the probability of C is $P(C) = n(C)/n(S) = 1/8 = 0.125$

d/ Compute the probability of having two girls & one boy in three child family.

For the event $D = \text{"two girls \& one boy"} = \{BGG, GBG, GGB\}$, we have $n(D) = 3$, Since the outcome are equally likely, the probability of D is $P(D) = n(D)/n(S) = 3/8 = 0.375$.

Example: A box of 80 candles consists of 30 defective and 50 non defective candles. If 10 of these candles are selected at random without replacement, what is the probability?

a) All will be defective?

b) 6 will be non-defective?

c) All will be non-defective?

Solution

$$\text{Total Selection: } \binom{80}{10} = N = n(S)$$

a) Let A be the event that all will be defective.

Total way in which A occur = $\binom{30}{10} * \binom{50}{0} = NA = n(A)$

$$P(A) = \frac{n(A)}{n(S)} = \binom{30}{10} * \binom{50}{0} / \binom{80}{10} = 0.00001825$$

b) Let A be the event that 6 will be non-defective.

Total way in which A occur = $\binom{30}{4} * \binom{50}{6} = NA = n(A)$

$$P(A) = \frac{n(A)}{n(S)} = \binom{30}{4} * \binom{50}{6} / \binom{80}{10} = 0.265$$

c) Let A be the event that all will be non-defective.

Total way in which A occur = $\binom{30}{0} * \binom{50}{10} = NA = n(A)$

$$P(A) = \frac{n(A)}{n(S)} = \binom{30}{0} * \binom{50}{10} / \binom{80}{10} = 0.00624.$$

5.1.2. The Frequentist Approach (Empirical Probability)

This approach to probability is based on relative frequencies.

Definition: Suppose we do again and again a certain experiment n times and let A be an event of the experiment and let k be the number of times that event A occurs. Therefore the probability of the event A happening in the long run is given by:

$$P(A) = \frac{\text{Number of times event A has occurred}}{\text{Total number of observations}} = \frac{k}{n}$$

In other words given a frequency distribution, the probability of an event (A) being

$$\text{in a given class is } P(A) = \frac{\text{Frequency of class A}}{\text{Total frequency in the distribution}}$$

Example 5.16: The national center for health statistics reported that of every 539 deaths in recent years, 24 resulted that from automobile accident, 182 from cancer, and 353 from other disease. What is the probability that particular death is due to an automobile accident?

Solution

$$P(\text{automobile}) = \text{death due to automobile} / \text{total death} = 24/539 = 0.445$$

The probability that particular death is due to an automobile accident is 0.445.

5.1.3. The axiomatic approach

Let E be a random experiment and S be a sample space associated with E . With each event A a real number called the probability of A satisfies the following properties called axioms of probability or postulates of probability.

1. $0 \leq P(A) \leq 1$
2. $P(S) = 1$, S is the sure/certain event.
3. If A_1 and A_2 are mutually exclusive events, the probability that one or the other occur equals the sum of the two probabilities. i. e. $P(A_1 \cup A_2) = P(A_1) + P(A_2)$
Similarly $P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n) = \sum_{i=1}^n P(A_i)$
4. $P(A') = 1 - P(A)$
5. $P(\emptyset) = 0$, $\emptyset$ is the impossible event.

Some probability rules (Derived theorems of probability)

Rule 1: let A be an event and A' be the complement of A with respect to a given sample space of an experiment, then $P(A') = 1 - P(A)$

Proof: let S be a sample space $S = A \cup A'$ and, A and A' are mutually exclusive

$$A \cap A' = \emptyset$$

$$P(S) = P(A \cup A') = P(A') + P(A) \text{ and } P(S) = 1$$

$$1 = P(A') + P(A) \Rightarrow P(A') = 1 - P(A)$$

Rule 2: let A and B are events of a sample space S , then

$$P(A' \cap B) = P(B) - P(A \cap B)$$

$$\textbf{Proof: } B = S \cap B = (A \cup A') \cap B = (A \cap B) \cup (A' \cap B)$$

If $A \cap B \neq \emptyset$, then $P(B) = P(A \cap B) + P(A' \cap B)$

$$P(A' \cap B) = P(B) - P(A \cap B).$$

Rule 3: Suppose A and B are two events of a sample space, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof:

$$(A \cup B) = A \cup (A' \cap B), \text{ } A \text{ and } A' \cap B \text{ are disjoint sets}$$

$$\therefore P(A \cup B) = P(A) + P(A' \cap B) \dots (*)$$

But we have already proved that $P(A' \cap B) = P(B) - P(A \cap B)$

Put this in equation (*)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example: A fair die is thrown twice. Calculate the probability that the sum of spots on the face of the die that turn up is divisible by 2 or 3.

Solution

$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$

This sample space has $6 \times 6 = 36$ elements let A be the event that the sum of the spots on the die is divisible by 2 and B be the event that the sum of the spots on the die is divisible by three, then

$A = \{(1,1), (1,3), (1,5), (2,2), (2,4), (2,6), (3,1), (3,3), (3,5), (4,2), (4,4), (4,6), (5,1), (5,3), (5,5), (6,2), (6,4), (6,6)\}$

$B = \{(1,2), (1,5), (2,1), (2,4), (3,3), (3,6), (4,2), (4,5), (5,1), (5,4), (6,3), (6,6)\}$

$A \cap B = \{(1, 5), (2, 4), (3,3), (4,2), (5,1), (6,6)\}$

$P(A \text{ or } B) = P(A \cup B)$

$= P(A) + P(B) - P(A \cap B)$

$= 18/36 + 12/36 - 6/36 = 24/36 = 2/3$

Exercise -5

1. In how many ways can 10 people be seated on a bench if only 4 seats are available?
2. A committee of 5 is formed by drawing lots from 8 boys and 6 girls. Find the probability that the committee will consist of 2 boys and 3 girls.
3. A box contains 6 red, 4 white and 5 black balls. A man draws 4 balls from the box at random. Find the probability that among the balls drawn there is at least one ball of each color.
4. How many four-digit numbers can be formed with the 6 digits 1,2, . . .,6, if
a/ repetitions are allowed b/repetitions are not allowed.
5. The probabilities that A and B solve a given problem independently are $2/3$ and $3/5$ respectively. If both of them attempt the problem, find the probability that the problem will be solved.

Chapter-7

7. Probability Distribution

7.1 the Concept of Random Variables

Before introducing a probability distribution, we have to define what is meant by a random variable, and provide some brief demonstrations based on finite sample spaces.

Definition 7.1

A random variable is defined as variable (typically represented by capital letters such as X) whose values are determined by chance with associated probabilities, for each outcome of a procedure.

For instance, we can define X based on possible genotypes of a bi-allelic gene A as follows:

$$X = \begin{cases} 0 & \text{for genotype } AA, \\ 1 & \text{for genotype } Aa, \\ 2 & \text{for genotype } aa. \end{cases}$$

A random variable can be discrete or continuous.

Definition 7.1.1

Discrete Random Variable: If the random variable X can assume only a particular finite or countably infinite set of values, it is said to be a discrete random variable. For example, if you throw a die, the outcome X is a random variable, which can assume only the values 1, 2, 3, 4, 5 and 6.

“Countable” refers to the fact that there might be infinitely many values, but they can be associated with the counting process.

Discrete random variables can be categorical (nominal or ordinal), such as genotype, or counts, such as the number of visiting an emergency room per day,

Examples discrete random variables

- Number of heads if a coin is flipped 3 times

- Number of girls in 14 new born babies
- Genotypes of a bi-allelic gene A

Definition 7.1.2

Continuous Random Variable: A random variable X is said to be continuous if it can take all possible values (integral as well as fractional) between certain limits.

Continuous random variables occur when we deal with quantities that are measured on a continuous scale. For instance, the life length of an electric bulb, the speed of a car, weights, heights, and the like are continuous. In such cases, probabilities are associated with intervals or regions of a continuous random variable, and not with individual points.

Examples of continuous random variable

- The height of an individual
- The distance between Bahir Dar and Addis Ababa
- The weight of an individual etc.

Definition 7.2

A probability distribution is a graph, table or an equation (formula) that links each outcome of a statistical experiment with its probability of occurrence.

Probability distributions can be either discrete or continuous probability distributions.

7.2.1 Discrete probability distributions

For discrete random variables, the probability distribution is fully defined by the probability mass function (pmf) is usually denoted by $p(x)$. This is a function that specifies the probability of each possible value within range of random variable.

If X is a discrete random variable taking at most a countably infinite number of values $x_1, x_2, \dots$, then $P(x_i) = P(X = x_i)$; $i = 1, 2, \dots$ is called the probability mass function of random variable X . The set of ordered pairs $\{x_i, P(x_i)\}$ $i = 1, 2, \dots$ gives the probability distribution of the random variable X . The numbers $P(x_i)$; $i = 1, 2, \dots$ must satisfy the following conditions.

For discrete random variables, the pmf provides the probability of each possible value.

$$P(x_i) \geq 0$$

$$\sum_{i=1}^{\infty} P(X = x_i) = 1$$

For discrete random variable, $P(X \leq x_1) \neq P(X < x_1)$ and $P(X \geq x_1) \neq P(X > x_1)$

Example 7.1 Consider an experiment of "flipping a fair coin 3 times". List the elements of the sample space that are assumed to be equally likely (as this is what is meant by a fair or balanced coin) and the corresponding values x of the r-v X , the number of heads observed.

Solution: If H stands for heads and T for tails, then the sample space corresponding to this experiments is $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$.

Since X = the number of heads observed, the results are shown in the following table:

Element of sample space	Probability	x
HHH	1/8	3
HHT	1/8	2
HTH	1/8	2
HTT	1/8	1
THH	1/8	2
THT	1/8	1
TTH	1/8	1
TTT	1/8	0

Thus, we can write $X(HHH) = 3$, $X(HHT) = 2$, ..., $X(TTT) = 0$, and $P(X = 3) = 1/8$ = the probability that the r-v X is 3, $P(X = 2) = 3/8$, and $P(X=0)=1/8$.

Note the possible values of X are 0,1,2 and 3.

$P(X = 0) = P(X (TTT)) = 1/8$, $P(X=1) = P(X (HTT))+P(X (THT))+ P(X (TTH))=1/8+1/8+1/8 = 3/8$, $(X=2) = P(X (HHT)) +P(X (HTH)) +P(X (THH)) = 1/8+1/8+1/8 = 3/8$, $P(X=3) = P(X (HHH)) = 1/8$.

The probability distribution for the above example is provided by the following table (Table 6.1).

Table 7.1 Probability distribution

X (number of heads)	0	1	2	3
P(X=x)	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Example 7.2: A study consists of randomly selecting 14 new born babies and counting the number of female newborns. If we assume that girls and boys are equally likely and if we let X denote number of new born girls among 14 new born then X is a random variable because its value depends on chance. The possible values of X are 0, 1, 2, . . . , 14. Table 6.2 lists the value of X along with the corresponding probabilities.

Table 6.2 Probability distribution

Number of girls (X)	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
P(X=x)	0.0000	0.0001	0.0006	0.0022	0.0061	0.0122	0.0183	0.0209	0.0183	0.0122	0.0061	0.0022	0.0006	0.0001	0.0000

7.2.2 Continuous probability distributions

Continuous probability distribution is a probability distribution whose random variable is continuous. It describes an “unbroken” continuum of possible occurrences. For continuous random variables, we use probability density functions (pdf) usually denoted by $f(x)$, to specify the distribution. Using the pdf, we can obtain the probability of any interval.

A random variable, X , is said to be a continuous random variable if there is a non-negative

function, f , such that $P(X \leq x) = F(x) = \int_{t=-\infty}^x f(t) dt$

The function of X , $f(x)$, it satisfies the following conditions.

i. $f(x) \geq 0$ for all x , $-\infty < x < \infty$

ii. The total area under the probability density curve is 1, that is $\int_{-\infty}^{\infty} f(x) dx = 1$

iii. For continuous random variables, the number of possible values is uncountable, and the probability of any specific (single) value is zero.

iv. If we are interested in the probability that the value of the random variable is within a specific interval from x_1 to x_2 ; we show this probability as $P(x_1 < X \leq x_2) = P(x_1 \leq X \leq x_2) = P(x_1 \leq X < x_2) = P(x_1 < X < x_2)$, where $x_1 < x_2$ and this is the area bounded by curve of probability density function and interval x_1 and x_2 on x -axis. That is:

$$P(x_1 < X \leq x_2) = P(x_1 \leq X \leq x_2) = P(x_1 \leq X < x_2) = P(x_1 < X < x_2) = \int_{x_1}^{x_2} f(x) dx$$

v. The probability of any interval from x_1 to x_2 , where $x_1 < x_2$, can be obtained using the corresponding lower tail probabilities for these two points as follows: $P(x_1 \leq X \leq x_2) = P(X \leq x_2) - P(X \leq x_1)$

vi. The integration from x_1 to x_2 in the case of the continuous variable is analogous to the summation of probabilities in the discrete case.

Example 6.4: A continuous random variable X has a probability density function given by

$$f(x) = \frac{1}{4}x + \frac{1}{2}, 0 \leq X \leq 1.$$

Find the probability that X lies between the interval 0 and 1.

$$\text{Solution: } \int_0^1 \left(\frac{1}{4}x + \frac{1}{2} \right) dx = \frac{1}{8}x^2 + \frac{1}{2}x \Big|_0^1 = \frac{1}{8} + \frac{1}{2} = \frac{5}{8}$$

7.2.3 Expectation and Variance of Random variable

The objective of this section is to introduce you with the most common parameters of probability distributions. There are some summary measures in terms of which we can summarize the behavior of probability distributions. The most common of these are the average called *expected value* and dispersion about the average called *the variance*.

7.2.4. Expectation

The averaging process, when applied to a random variable is called expectation. It is denoted by $E(X)$ or μ and is read as the expected value of X or the mean value of X .

Case 1: For discrete random variable

Suppose X is a discrete random variable which takes on values in a finite set $x_1, x_2, \dots, x_n$ with probabilities $P(x_i) = P[X = x_i]$ $i = 1, 2, \dots, n$, then the expected value of X , $E(X)$ of the discrete random variable is given by:

$$E(x) = \mu = \sum_{i=1}^n x_i P(x_i)$$

Case 2: For continuous random variable

If X is a continuous random variable then

$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$ provided $\int_{-\infty}^{\infty} |x| f_X(x) dx < \infty$ where $f_X(x)$ is the probability density function of the continuous random variable X .

Case 3: Mathematical expectation of some real function $h(x)$ of a discrete random variable is given by:

$$E[h(x)] = \sum_{i=1}^n h(x_i) P(x_i)$$

Similarly if X is a continuous random variable, then

$$E[h(x)] = \int_{-\infty}^{\infty} h(x) f_X(x) dx$$

Properties of Expectation

If X and Y are random variables and a, b are constants then:

1. $E(k) = k$, where k is any constant

2. $E(kX) = k E(X)$, where k is any constant
3. $E(X + k) = E(X) + k$
4. $E(X + Y) = E(X) + E(Y)$
5. $E(XY) = E(X) E(Y)$, if X, Y are independent random variables
6. $E(X) \geq 0$, if $X \geq 0$.
7. $|E(X)| \leq E(|X|)$
8. $|E(XY)| \leq E(X^2) E(Y^2)$.

7.3.2 Mean and variance of a random variables

Mean of $X = E(X)$

$$\text{Variance of } X = \sigma_x^2 = E(X^2) - [E(X)]^2 = E[X - E(X)]^2$$

Case 1: If X is a discrete random variable with expected value μ then the variance of X , denoted by $\text{Var}(X)$, is defined by:

$$\sigma_x^2 = \text{Var}(X) = E(X - \mu)^2 = E(X^2) - \mu^2 = \sum_{i=1}^n (x_i)^2 P(x_i) - \mu^2$$

$$\text{Alternatively, } \text{Var}(X) = \sum_{i=1}^n (x_i - \mu_x)^2 P(x_i)$$

Case 2: If X is a continuous random variable, then $\text{var}(X), \sigma_x^2 = \int_{-\infty}^{\infty} (x - \bar{x})^2 f_x(x) dx$

Properties of Variances

- a. Scaling and shifting random variables: For any random variable X and constant $\mathbf{a}$, it can be shown that

$$\text{Var}(\mathbf{a}X) = \mathbf{a}^2 \text{Var}(X)$$

$$\text{Var}(X + \mathbf{a}) = \text{Var}(X) + 0 = \text{Var}(X)$$

- b. If X and Y are independent random variables, then $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$

More generally if $X_1, X_2, \dots, X_k$ are independent random variables, then

$\text{Var}(X_1 + X_2 + \dots + X_k) = \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_k)$, i.e.

$$\text{Var}\left(\sum_{i=1}^k X_i\right) = \sum_{i=1}^k \text{Var}(X_i)$$

c. If X and Y are *not* independent, then $\text{Var}(X+Y) = \text{Var}(X) + 2\text{Cov}(X,Y) + \text{Var}(Y)$ and $\text{Var}(X-Y) = \text{Var}(X) - 2\text{Cov}(X,Y) + \text{Var}(Y)$

Example 6.5: Two fair coins are tossed. Determine $\text{Var}(X)$ where X is the number of heads that appear.

a) Use the definition of the variance.

b) Use the fact that the variance of the sum of independent variables is equal to the sum of the variance.

Solution:

a) Let X is number of heads with possible values 0,1 and 2. The Sample space consists of {HH, TH, HT, TT}

$$P(X=0) = 1/4, P(X=1) = 1/2, P(X=2) = 1/4$$

$$E(X) = 0 \cdot P(X=0) + 1 \cdot P(X=1) + 2 \cdot P(X=2)$$

$$= 0(1/4) + 1(1/2) + 2(1/4)$$

$$= 1.$$

$$E(X^2) = 0^2 P(X=0) + 1^2 \cdot P(X=1) + 2^2 P(X=2)$$

$$= 0(1/4) + 1(1/2) + 4(1/4)$$

$$= 3/2.$$

Implies that, $\text{Var}(X) = E(X^2) - \mu^2 = 3/2 - 1 = 1/2$

b) Let X be head on the first coin with possible values 0 and 1

Y be head on the second coin with possible values 0 and 1.

$$P(X=0) = 1/2, P(X=1) = 1/2 \quad \text{and} \quad P(Y=0) = 1/2, P(Y=1) = 1/2$$

$$E(X) = 0.P(X=0) + 1.P(X=1) = 0(1/2) + 1(1/2) = 0(1/2) + 1(1/2) = 1/2$$

$$E(Y) = 0.P(Y=0) + 1.P(Y=1) = 1/2$$

$$E(X^2) = 0^2.P(X=0) + 1^2.P(X=1) = 0(1/2) + 1(1/2) = 1/2$$

$$E(Y^2) = 0^2.P(Y=0) + 1^2.P(Y=1) = 0(1/2) + 1(1/2) = 1/2$$

$$\text{Var}(X) = E(X^2) - \mu^2 = 1/2 - (1/2)^2 = 1/4$$

$$\text{Var}(Y) = E(Y^2) - \mu^2 = 1/2 - (1/2)^2 = 1/4$$

X and Y are independent (i.e. the outcome of one coin does not influence the outcome of the second)

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) = 1/4 + 1/4 = 1/2.$$

Example 6.6: Compute the variance of $f(x) = \frac{x^2}{9}$ for $0 < x < 3$

$$V(x) = E(x^2) - [E(x)]^2$$

$$E(x^2) = \int_0^3 x^2 \left(\frac{x^2}{9} \right) dx = \int_0^3 \frac{x^4}{9} dx = \frac{1}{9} \left[\frac{x^5}{5} \right]_0^3 = \frac{27}{5}$$

$$E(x) = \int_0^3 x \left(\frac{x^2}{9} \right) dx = \frac{1}{9} \left[\frac{x^4}{4} \right]_0^3 = \frac{9}{4}$$

$$\text{Therefore, } V(x) = E(x^2) - [E(x)]^2$$

$$= \frac{27}{5} - \left(\frac{9}{4} \right)^2 = \underline{\underline{0.34}}$$

7.3 Common Discrete Probability Distributions

7.3.1 Binomial Distribution

The origin of binomial distribution is Bernoulli's trial. Bernoulli's trial is an experiment where there are only two possible outcomes, "success" or "failure". In connection with this trial, a success may be getting heads with a balanced coin; it may be passing an examination. Whenever

we face such experiment, we use binomial distribution under the assumptions stated below. Any experiment can also be turned into a Bernoulli trial by defining one or more possible results which we are interested as “Success” and all other possible results as “Failure”. For instance, while rolling a fair die, a "success" may be defined as "getting even numbers on top" and odd numbers as "Failure".

Generally, the sample space in a Bernoulli trial is $S = \{S, F\}$, S = Success, F = failure.

Notation: Let probability of success and failure are p and q respectively.

$P(\text{success}) = P(s) = p$ and $P(\text{failure}) = P(f) = q$, where $q = 1 - p$.

Definition: Let X be the number of success in n repeated Bernoulli trials with probability of success p on each trial, then the probability distribution of a discrete random variable X is called binomial distribution. Let p = the probability of success $q = 1 - p$ = the probability of failure on any given trial. A binomial random variable with parameters n and p represents the number of r successes in n independent trials, when each trial has p probability of success.

If X is a random variable, then for $r = 0, 1, 2, \dots, n$

$$P(X = r) = \frac{n!}{r!(n-r)!} p^r (1-p)^{n-r} = \frac{n!}{r!(n-r)!} p^r q^{n-r}, \text{ where } q = 1 - p$$

A binomial experiment is a probability experiment that satisfies the following assumptions.

The experiment consists of n identical trials.

Each trial has only one of the two possible mutually exclusive outcomes, success or a failure.

The probability of each outcome does not change from trial to trial.

The trials are independent.

Examples of random variables that usually obey the Binomial distribution are

- ✓ Blood samples are tested for the presence or absence of an antigen
- ✓ Rates fed a potential carcinogen are examined for tumors
- ✓ People classified as having or not having cleft lip

- ✓ Newborn children are classified as having or not having Down syndrome

If X is a binomial random variable with two parameters n and p then

$$\text{i) } E(X) = np$$

$$\text{ii) } \text{Var}(X) = npq$$

Example 6.7: A fair coin is flipped 3 times, what is the probability of getting exactly two heads?

Solution:

Let X be number of heads with possible values 0,1,2,3

$P(\text{getting head}) = p = \frac{1}{2}$, $q = 1-p = \frac{1}{2}$, $n = 3$

$$P(X = 2) = \frac{3!}{2!((3-2)!)} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{3-2} = \frac{3}{8}$$

Example 6.8: Ten patients are treated surgically. For each person there is 70% of being successful surgery. What is the probability of only five or fewer successful surgeries.

Solution

$n=10$, $p=0.7$, $q=1-p=0.3$

$$P(X \leq 5) = P(X = 5) + P(X = 4) + P(X = 3) + P(X = 2) + P(X = 1) + P(X = 0)$$

$$= \binom{10}{5} (0.7)^5 (0.3)^{10-5} + \binom{10}{4} (0.7)^4 (0.3)^{10-4} + \dots + \binom{10}{0} (0.7)^0 (0.3)^{10-0}$$

$$0.1029 + 0.0368 + 0.009 + 0.0014 + 0.0001 + 0.0000 = 0.1502$$

7.3.2 Poisson Distribution

Another important discrete probability distribution is the *Poisson distribution*. It is a discrete probability distribution which is used in the area of rare events. The Poisson distribution counts the number of success in a fixed interval of time or within a specified region.

Examples of random variables that usually obey the Poisson distribution are:

- The number of car accidents in a day.
- The chance of a cavalryman being killed by a horse kick in the course of a year

- Arrival of telephone calls over interval of times.
- The number of misprints on a typed page (a group of pages) of a book.
- Natural disasters like earth quake.
- The number of suicides reported by a particular city.
- The number of customers entering a post office on a given day.

To apply the Poisson distribution, two conditions must be met:

- ✓ The number of success that occurs in any interval is independent of those that occur in other non-overlapping intervals.
- ✓ The probability of a success in an interval is proportional to the size of the interval. In short, the two important traits of the Poisson distribution are *independence* and *probability*.

Let X is the number of occurrences in a Poisson process and λ be the actual average number of occurrence of an event in a unit length of interval, the probability function for Poisson distribution is,

$$P((X)) = \frac{\lambda^x e^{-\lambda}}{x!}, x = 0, 1, 2, \dots$$

Remarks

Poisson distribution possesses only one parameter λ

If X has a Poisson distribution with parameter λ , then $E(X) = \lambda$ and $\text{Var}(X) = \lambda$,

i.e. $E(X) = \text{Var}(X) = \lambda$,

$$\sum_{i=0}^{\infty} P(x_i) = 1$$

Example 6.9 In a small city, 10 accidents took place in a time of 50 days. Find the probability that there will be a) two accidents in a day and b) three or more accidents in a day.

Solution:

There are 0.2 accidents per day.

Let X be the random variable, the number of accidents per day

$X \sim \text{poiss} (\lambda = 0.2)$ $X = 0, 1, 2, \dots$

$$P((X = 2)) = \frac{(0.2)^2 e^{-0.2}}{2!} = 0.0164$$

$$\text{b) } P(X \geq 3) = P(X = 3) + P(X = 4) + P(X = 5) + \dots$$

$$= 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

$$\dots\dots\dots \text{since } \sum_{i=0}^{\infty} P(x_i) = 1$$

$$= 1 - [0.8187 + 0.1637 + 0.0164] = 0.0012$$

7.4 Common Continuous Probability Distributions

7.4.1 Normal Distributions

It is the most important distribution in describing a continuous random variable and used as an approximation of other distribution. Many variables in the practical world follow this distribution, and hence in many ways it is the cornerstone of modern Statistical Theory. It has been noticed that empirical distributions of various types of observations in natural and social sciences are often very close to normal distribution. In statistical analysis the distributions of observations is frequently assumed to be approximately normal. In statistical estimation and testing of hypotheses the normal distribution plays an important role.

A random variable X has a normal distribution with parameters μ & σ^2 and it is known as a normal random variable if its pdf is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\} = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \text{ where } -\infty < \mu < \infty, -\infty < x < \infty \text{ \& } \sigma > 0.$$

The graph of the normal distribution is known as the normal curve, which is bell-shaped:



Figure Normal probability curve

SOME PROPERTIES OF THE NORMAL CURVE

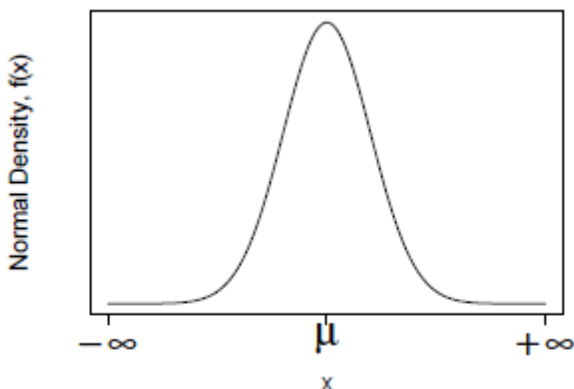
The following are the important properties of the normal curve:

1. The normal probability density function for $X \sim N(\mu, \sigma^2)$ is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty \text{ where } \pi = 3.14 \text{ and } e = 2.72 \text{ are mathematical constants}$$

and μ (mean) and σ (standard deviation) are parameters of normal distribution

2. The normal curve is “bell-shaped” and symmetrical about the mean. The property of symmetry can be shown using the pdf as: $f(\mu + c) = f(\mu - c)$. This is equivalent to saying that $P(X \leq \mu) = P(X \geq \mu) = 0.5$



3. Since this is the property of the median, it follows that, for the normal distribution, Mean = Median = Mode.

4. The height of the normal curve is at its maximum when $X = \mu = \text{mean}$, which means, again, Mean = Median=Mode. This property can also be verified using the first and second derivative tests; that is, $f'(x) = 0 \Rightarrow x = \mu$. This shows $x = \mu$ may be maximum or minimum value of X , but using the second derivative test, $f''(\mu) = \frac{-1}{\sigma^3 \sqrt{2\pi}} < 0$, we see that the point is the maximum value. Therefore, by property 2 and 3, we can conclude that, the mean, median and mode coincide for the normal curve.
5. The normal curve is asymptotic to the X - axis.
6. The first and the third quartiles are equidistant from the median,
i.e., $Q_3 - Q_2 = Q_2 - Q_1$. Or, $Q_2 = \frac{Q_1 + Q_3}{2}$.
7. The Probability that a random variable will have a value between any two points is equal to the area under the curve between those points.
8. Area under curve equals 1

6.4.2. Standard Normal Distribution

The symmetrical property of the normal distribution provides a means that is helpful in calculating probabilities, which is also facilitated by transforming any normal distribution with any mean and variance to the standard normal distribution.

By standardization we mean that the random variable X will be transformed to another random variable whose mean is 0 and variance is 1. The normal distribution with zero mean and standard deviation one is known as *standard normal distribution*. If X has normal distribution with mean μ_x and standard deviation σ , then the standard normal distribution Z is given by

$$Z = \frac{x - \mu}{\sigma}, \text{ for population}$$

$$Z = \frac{x - \bar{x}}{s}, \text{ for sample}$$

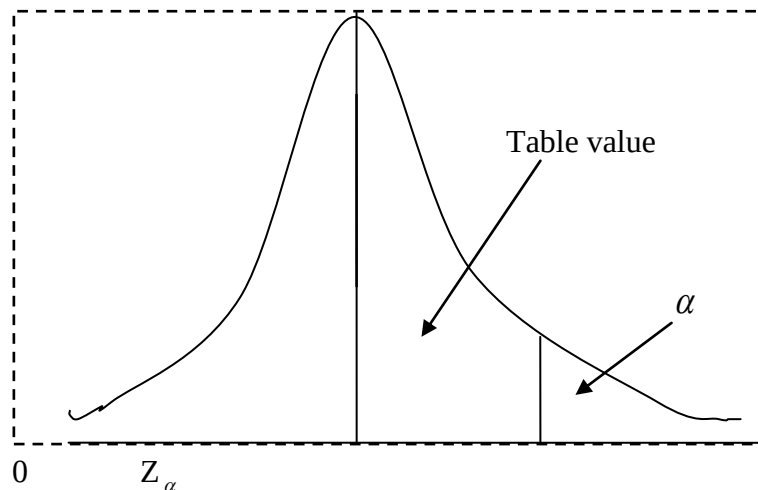
Using the properties of expectations, it is now trivial to show that $E(Z) = 0$ and $V(Z) = 1$.

The pdf of Z is, thus, given by $f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$, $-\infty < z < \infty$.

The entries in Table A of the Appendix are the values of $P(0 < Z < z) = \int_0^z f(z) dz$.

That is, the table gives us the probabilities that a random variable Z having the standard normal distribution will take on a value on the interval from 0 to z, for $z = 0.00, 0.01, 0.02, \dots, 3.98$, and 3.99; due to the symmetrical property of the normal curve with respect to its mean, it is unnecessary to extend the table for negative values of Z.

Note that $P(Z < 0) = P(Z > 0) = 0.5$.



Tabulated areas under the standard N.D from 0 to z

That is, the arrowed region is called table value which is given by:

$$P(0 < Z < z) = \int_0^z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz.$$

Basic Properties of the standard normal Curve:

1. Total area under the standard normal curve is equal to 1.
2. The standard normal curve is asymptotic to x-axis.
3. The standard normal curve is symmetric about 0.
4. Most of the area under the standard normal curve lies between $z = -3$ and $z = 3$.

Given a normal distributed random variable X with mean μ and standard deviation σ

$$P(a < X < b) = P\left(\frac{a-\mu}{\sigma} < \frac{x-\mu}{\sigma} < \frac{b-\mu}{\sigma}\right)$$

$$P(X < a) = P\left(\frac{x-\mu}{\sigma} < \frac{a-\mu}{\sigma}\right)$$

$$\text{But, } \frac{x-\mu}{\sigma} = Z \text{ standard normal random variable } P\left(Z < \frac{a-\mu}{\sigma}\right)$$

$$\text{Note: i) } P(a < x < b) = P(a \leq X < b)$$

$$= P(a < X \leq b)$$

$$= P(a \leq X \leq b)$$

$$\text{ii) } P(-\infty < Z < \infty) = 1$$

Example 6.10: Find the probabilities that a random variable having the standard normal distribution will take on a value

- a. Less than 1.72;
- b. Less than -0.88;
- c. Between 1.30 and 1.75;
- d. Between -0.25 and 0.45.

a) $P(Z < 1.72) = P(Z < 0) + P(0 < Z < 1.72) = 0.5 + 0.4573 = 0.9573$.

b) $P(Z < -0.88) = P(Z > 0.88) = 0.5 - P(0 < Z < 0.88) = 0.5 - 0.3106 = 0.1894$.

c) $P(1.30 < Z < 1.75) = P(0 < Z < 1.75) - P(0 < Z < 1.30) = 0.4599 - 0.4032 = 0.0567$.

$$\begin{aligned} \text{d) } P(-0.25 < Z < 0.45) &= P(-0.25 < Z < 0) + P(0 < Z < 0.45) = P(0 < Z < 0.25) + P(0 < Z < 0.45) \\ &= 0.0987 + 0.1736 = 0.2723. \end{aligned}$$

Application of the Standard Normal Distribution

Let $X \sim N(\mu, \sigma^2)$. Suppose that we want to find the probability $P(a < X < b)$.

Since a , b , μ , and σ are known (given), we standardize a , b and X as:

$$P(a < X < b) = P\left(\frac{a - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{b - \mu}{\sigma}\right) = P(z_1 < Z < z_2), \text{ say.}$$

Now, we need only to get the readings from the Z- table corresponding to z_1 and z_2 to get the required probabilities, as we have done in the preceding example.

Also, we can find the following one-sided probabilities:

$$P(X < b) = P\left(Z < \frac{b - \mu}{\sigma}\right) = P(Z < z_2), \text{ and } P(X > a) = P\left(Z > \frac{a - \mu}{\sigma}\right) = P(Z > z_1).$$

We have seen that a Z- value measures the distance between a particular value of X and the mean in units of standard deviation.

Example 6.10: If $X \sim N(\mu, \sigma^2)$, find the probabilities

$$\text{a) } P(\mu - \sigma < X < \mu + \sigma); \text{ b) } P(\mu - 2\sigma < X < \mu + 2\sigma); \text{ c) } P(\mu - 3\sigma < X < \mu + 3\sigma).$$

Solution: As in the case of $P(a < X < b)$, we simply replace a and b .

$$\begin{aligned} \text{a) } P(\mu - \sigma < X < \mu + \sigma) &= P\left(\frac{\mu - \sigma - \mu}{\sigma} < Z < \frac{\mu + \sigma - \mu}{\sigma}\right) \\ &= P(-1 < Z < 1) = 2P(0 < Z < 1) = 2(0.3413) \text{ (See Table A)} \\ &= 0.6828 \text{ or } 68.28\%. \end{aligned}$$

$$\text{b) Similarly, } P(\mu - 2\sigma < X < \mu + 2\sigma) = P(-2 < Z < 2) = 2P(0 < Z < 2) = 2(0.4772) = 0.9544.$$

c)

$$P(\mu - 3\sigma < X < \mu + 3\sigma) = P(-3 < Z < 3) = 2P(0 < Z < 3) = 2(0.4987) = 0.9974, \text{ or } 99.74\%.$$

From which we can tell that, about 68.30% lies in the region $\mu - \sigma$ & $\mu + \sigma$ (1 Standard deviation on either side).

About 95.50% lies in $\mu - 2\sigma$ & $\mu + 2\sigma$ (2 Standard Deviations on either side).

About 99.7% lies in $\mu - 3\sigma$ & $\mu + 3\sigma$ (3 Standard Deviations on either side).

Notation: z_α denotes the value of Z for which the area to its right is equal to α .

This notation is useful in statistical inference, and note that finding z_α is identical with reading anti-logarithms.

Example 6.11: Find a) $z_{0.01}$; b) $z_{0.05}$

Solution: a) $z_{0.01}$ corresponds to an entry of $0.5 - 0.01 = 0.4900$.

In Table A, look for the value closest to 0.4900, which is 0.4901, and the Z value for this is $Z = 2.33$. Thus, $z_{0.01} = 2.33$.

b) Again, $z_{0.05}$ is obtained as $0.5 - 0.05 = 0.4500$, which lies exactly between 0.4495 and 0.4505, corresponding to $Z = 1.64$ and $Z = 1.65$. Hence, using interpolation, $z_{0.05} = 1.645$.

Example 6.12: Suppose that $X \sim N(165, 9)$, where X = the breaking strength of cotton fabric. A sample is defective if $X < 162$. Find the probability that a randomly chosen fabric will be defective.

Solution: Given that $\mu = 165$ and $\sigma^2 = 9$,

$$P(X < 162) = P\left(\frac{X - \mu}{\sigma} < \frac{162 - \mu}{\sigma}\right) = P\left(Z < \frac{162 - 165}{3}\right)$$

$$= P(Z < -1) = 0.5 - P(-1 < Z < 0) \quad (\text{Since } P(Z < 0) = 0.5)$$

$$= 0.5 - P(0 < Z < 1) \quad (\text{By symmetry})$$

$$= 0.5 - 0.3413 = 0.1587 \quad (\text{Table value for } Z = 1)$$

Example 7.13: Suppose a borderline hypertensive is defined as a person whose diastolic blood pressure is between 90 and 95 mm Hg inclusive, and the subjects are 35-44-year-old males whose BP is normally distributed with mean 80 and variance 144. What is the probability that a randomly selected person from this population will be a borderline hypertensive?

Solution

Solution:

Let X represents diastolic blood pressure of randomly selected subject, then $X \sim N(80, 144)$

$$\begin{aligned} P(90 < X < 95) &= P\left(\frac{90-80}{12} < \frac{\bar{X}-\mu}{\sigma} < \frac{95-80}{12}\right) = P(0.83 < Z < 1.25) \\ &= P(Z < 1.25) - P(Z < 0.83) = 0.8944 - 0.7967 = 0.098 \end{aligned}$$

Thus, approximately 9.8% of this population will be borderline hypertensive.

7.4.3 Chi-Square Distribution

Chi-Square distribution may be derived from normal distributions, if X_i ($i = 1, 2, \dots, n$) are n independent normal varieties with mean μ_i and variance σ_i^2 ($i = 1, 2, \dots, n$) then

$\chi^2 = \sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i} \right)^2$ is a chi-square variate with n degrees of freedom. The probability density

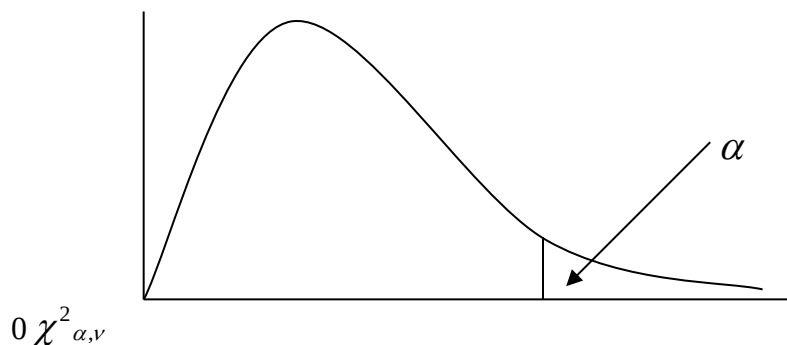
function of the χ^2 -distribution is given by

$$f(\chi^2) = \frac{1}{2^{n/2} \Gamma(n/2)} (\chi^2)^{(n/2-1)} e^{-\chi^2/2}, \quad 0 < \chi^2 < \infty \text{ where } n \text{ is the degree of freedom.}$$

Since the Chi-square distribution arises in many important applications, its values have been extensively tabulated. Table C at the end of this module contains values of $\chi^2_{\alpha, n}$ for $\alpha = 0.05, 0.025, 0.01, 0.005$ and $n = 1, 2, 3, \dots, 30$, where $\chi^2_{\alpha, n}$ is such that the area to its right under the Chi-square curve with n degrees of freedom is equal to α . That is, $\chi^2_{\alpha, n}$ is such that if X is a random variable having a Chi-square distribution with n degrees of freedom, then

$P(X \geq \chi^2_{\alpha, n}) = \alpha$. α is known as the level of significance. When n is greater than 30, the table

cannot be used and probabilities related to Chi-square distributions are usually approximated with normal distributions.



Properties of Chi-square Distribution

1. The exact shape of the distribution depends upon the number of degrees of freedom n . In general, when n is small, the shape of the curve is skewed to the right and as n gets larger, the distribution becomes more and more symmetrical.
2. The mean and variance of the χ^2 distribution are n and $2n$ respectively.
3. As $n \rightarrow \infty$ the χ^2 distribution approaches a normal distribution.
4. The sum of independent χ^2 varieties is also χ^2 variety.

7.4.4 The t-distribution

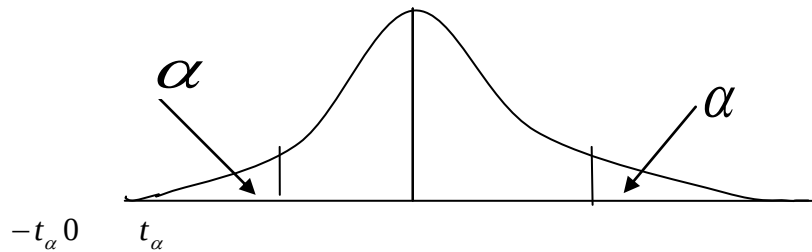
Let $X_1, X_2, \dots, X_n$ be a random sample drawn from a normal distribution having mean μ and standard deviation σ (unknown but estimated by S , sample standard deviation).

The statistic $t = \frac{\bar{X} - \mu}{S/\sqrt{n}}$ has t – distribution with $(n-1)$ degree of freedom where $\bar{X}$ is sample mean

and S is standard deviation.

In view of its importance, the t distribution has been tabulated extensively. Table B at the end of this module contains values of $t_{\alpha, n-1}$, for $\alpha = 0.10, 0.05, 0.025, 0.01, 0.005$, and $n = 1, 2, 3, \dots, 29$ degrees of freedom; where $t_{\alpha, n-1}$ is such that the area to its right under the curve of the t distribution with $(n-1)$ degrees of freedom is equal to α .

Notation: $t_{\alpha, (n-1)}$ stands for a value of t with $(n-1)$ degree of freedom the right of which an area equal to α in reading the tabulated values.



Student's t Distribution

Note:

1. The table value does not contain values of $t_{\alpha, n-1}$ for $\alpha > 0.50$, since the curve is symmetrical about $t=0$ (like the normal distribution) we have, $t_{\alpha, n-1} = -t_{\alpha, n-1}$
2. When $(n-1) = 30$ or more, probabilities related to the t distribution are usually approximated with the use of normal distributions.

Example 6.13: For a t -distribution with $n=20$, find t_{α} values leaving an area of

- 0.05 to the right; c) 0.10 to the left;
 0.975 to the right; d) half of $\alpha = 0.01$ on either side.

Solution; referring to Table B with $(n-1) = 19$ df , we have

- a) $t_{0.05} = 1.725$; c) $-t_{0.10} = -1.328$.
 b) $t_{0.975} = 2.093$; d) $t_{\frac{\alpha}{2}} = t_{0.005} = 2.861$; & $-t_{0.005} = -2.861$

Applications of t Distribution

The t distribution has wide applications in Statistics, only some are listed below:

Test of population Mean (One-sample t-test)

When we are dealing with a random sample of size $n < 30$, from a normal population, when σ^2 is unknown, the t distribution with $n-1$ degrees of freedom, is used to test the hypothesis that the population mean μ equals a given value (say, μ_0), against the alternatives: $\mu \neq \mu_0$, or $\mu > \mu_0$, or $\mu < \mu_0$.

Then, we calculate $t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$, which is to be compared with the table value $t_{\frac{\alpha}{2}}$, or t_{α} with $n-1$ degrees of freedom.

Note: The assumptions underlying student's t-distribution for such tests are:

The parent population from which the sample is drawn is normal.

The sample observations are independent; that is, the sample is random.

The population standard deviation (σ) is unknown.

n is small; that is $n < 30$.

Example: 6.14: In 16 one-hour test runs, the gasoline consumption of an engine averaged 16.4 gallons with a standard deviation of 2.1 gallons. In order to test the claim that the average gasoline consumption of this engine is 12.0 gallons per hour, calculate the t value and $t_{\alpha, n-1}$, for $\alpha = 0.05$.

Solution: Substituting $n=16$, $\mu=12.0$, $\bar{X}=16.4$, and $S=2.1$ in the formula, we get

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}} = \frac{16.4 - 12.0}{2.1 / \sqrt{16}} = 8.38; \text{ and the table value for } n-1=15 \text{ is } t_{0.05, 15} = 1.753.$$

t-Test of the equality of two means (two-sample t-test)

In many problems of applied research, we are interested in hypotheses concerning the difference between the means of two population means, $\mu_1 - \mu_2 = 0$, or the equality of two means

($\mu_1 - \mu_2 = 0$). In such tests, the following are assumed:

The parent populations from which the samples have been drawn are normally distributed;

The two population variances are equal, though unknown: $\sigma_1^2 = \sigma_2^2 = \sigma^2$.

The two samples are random and independent of each other;

The sample sizes are small: n_1 and/or n_2 are <30 .

t-Test of correlation and regression coefficients

In a *normal* regression and correlation analyses, it is used to test:

if the population regression coefficient equals to a certain constant ($\beta = \beta_o$); &

if the population correlation coefficient is significantly different from zero.

Exercise - 6

1. From a lot containing 20 items, of which 5 are defective, 4 are chosen at random. Let X be the number of defectives found.
 - a) Write down the pmf of X .
 - b) Find the probability distribution of X .
 - c) Find $E(X)$ and $V(X)$.
2. On the average, five smokers pass a certain street corners every ten minutes, what is the probability that during a given 10 minutes the number of smokers passing will be
 - a. 6 or fewer
 - b. 7 or more
 - c. Exactly 8.....
3. Suppose that total carbohydrate intake in 12-14 year old males is normally distributed with mean 124 g/1000 cal and standard deviation of 20 g/1000 cal.
 - a) What percent of boys in this age range have carbohydrate intake above 140g/1000 cal?
 - b) What percent of boys in this age range have carbohydrate intake below 90g/1000 cal?

4. Assume that among diabetics the fasting blood level of glucose is approximately normally distributed with a mean of 105 mg per 100 ml and SD of 9 mg per 100 ml.
 - a) What proportions of diabetics have levels between 90 and 125 mg per 100 ml?
 - b) What proportions of diabetics have levels below 87.4 mg per 100ml?
 - a) What level cuts off the lower 10% of diabetics?
 - b) What are the two levels which encompass 95% of diabetics?
5. An allergist claims that 45% of the patients she tests are allergic to some type of weed. What is the probability that
 - a. Exactly 3 of her next 4 patients are allergic to weeds?
 - b. None of her next 4 patients are allergic to weeds?
6. The amount of bread X (in hundreds of kg) that a certain bakery is able to sell in a day is found to be a continuous r-v with a pdf given as below:

$$f(x) = \begin{cases} kx & , 0 \leq x < 5 \\ k(10 - x) & , 5 \leq x < 10 \\ 0 & , \text{otherwise} \end{cases}$$
 - a) Find k ;
 - b) Find the probability that the amount of bread that will be sold tomorrow is
 - i) More than 500kg, ii) between 250 and 750 kg;
 - c) Find the expected amount of bread to be sold in any day.
7. Explain why the following experiments are not Binomial
 - a. Rolling a die until a 6 appears.
 - b. Asking 20 people how old they are.
8. Find the value of Z if the area between $-Z$ and Z is a) 0.4038; b) 0.8812; c) 0.3410.
9. Of a large group of men, 5% are less than 60 inches in height and 40% are between 60 & 65 inches. Assuming a normal distribution, find the mean and standard deviation of heights

Chapter-8

8 Sampling and Sampling Distribution of Mean

8.1 Basic Concepts

i. Population

A population is the complete collection of individuals, objects or measurements for which inferences are to be made. The population represents the target of an investigation, and the objective of the investigation is to draw conclusions about the population and it should be defined on the basis of the objective of the study by the investigator.

Example 7.1:

- ✓ In Systolic blood pressure for Japanese men, aged 45–69, living in Japan. The “population” can be considered to be the collection of blood pressures of all Japanese men.
- ✓ In the experimental situation to assess the effectiveness of a new special diet for children with phenylketonuria (PKU), the diet is used with a set of PKU children and their IQs are measured when they reach 4 years of age. The population is all PKU children who could potentially be treated with the new diet
- ✓ All students of DMU.
- ✓ Population of households in a certain village.

ii. Sample

A sample from a population is the set of measurements that are actually collected in the course of an investigation. It should be selected using some predefined sampling technique in such a way that they represent the population very well.

iii. Sampling (elementary) unit

Sampling (elementary) unit is the ultimate unit to be sampled or elements of the population to be sampled.

Example 8.2

- ✓ If somebody studies economic status of the households, households is the sampling unit.

- ✓ If one studies performance of freshman students in some college, the student is the sampling unit.

iv. **Sampling frame**

Sampling frame is the list of all elements (sampling units) in a population.

Examples 8.3

- ✓ List of households of a certain city.
- ✓ List of students in the registrar office of the university.

v. **Parameter and Statistic**

Parameter and Statistic are basic terms in sampling theory. Parameter is a value calculated from the population. For instance population mean, population variance, population proportion is parameters. Statistic is a value calculated from a sample. Sample mean, sample variance, sample proportion, etc are statistics.

vi. **Sampling error**

Sampling error is a type of error that may arise due to inappropriate sampling techniques applied. A sampling error is the difference between a sample statistic and its corresponding parameter. We can make probabilistic statements about this sampling error only if we have a probability sample.

vii. **Non-sampling error**

In addition to sampling error, the sample estimate may be subject to other errors, sampling errors. Errors in observation, interview or measurement error, errors due to non-response and errors in data processing: editing, coding, etc. The non-sampling error is likely to increase with increase in sample size. For instance a census survey may have non-sampling errors in large amount collected in the course of an investigation. It should be selected using some predefined sampling technique in such a way that they represent the population very well.

8.2 Reasons for sampling

Sample survey saves money:- It is possible to collect information from sample households and obtain estimates that reasonably approximate the actual characteristics of a large population. It obviously cheaper to gather information from 100 households rather than from 10,000 households.

Sample Survey saves time:-sample survey requires a smaller scale of operations at all stage and it reduces data collection and processing time.

Sample survey provides higher level of accuracy:-This accuracy can be achieved through more selective recruiting of interviewers and supervisors, more extensive training programs, a closer supervision of the personnel involved and a more efficient monitoring of the field work.

Sample survey could be the only option for the study in some specialized area. For example, there are some cases where information of technical nature requires highly trained personnel and specialized equipment like in medical areas.

Experimentation could be destructive in nature like testing industrial products such as testing the average duration of burning of bulbs, and testing the quality of wine, beer, etc. In this case sampling is the only feasible means of study.

8.3 Sampling Techniques

The technique of selecting a sample is important in sampling theory and usually it depends upon the nature of the investigation. The commonly used sampling techniques may be broadly classified as: Non Probability (Non-random) and Random (Probability) Sampling.

8.3.1 Random Sampling or probability sampling

Probability sampling techniques is a method of sampling in which all elements in the population have a pre-assigned probability to be included in to the sample.

In this sub-section, four different techniques of taking a random sample are discussed.

- a. Simple random sampling
- b. Stratified random sampling
- c. Cluster sampling
- d. Systematic sampling

a. Simple Random Sampling

In statistics, a simple random sample from a population is a sample chosen randomly, so that each possible sample has the same probability of being chosen. One consequence is that each member of the population has the same probability of being chosen as any other. In small

populations such sampling is typically done "without replacement", i.e., one deliberately avoids choosing any member of the population more than once. Although simple random sampling can be conducted with replacement, this is less common and would normally be described more fully as simple random sampling without replacement. Simple random sampling is a method of selecting n units out of a finite population of size N by giving equal probability to all units, or a sampling procedure in which all possible combinations of n units that may be formed from the finite population of size N units have the same probability of selection. There are ${}_N C_n$ distinct possible samples in the case of sampling without replacement; the chance of selecting each one of them is $\frac{1}{{}_N C_n}$. There are N^n possible samples in the case of sampling with replacement, the chance of selecting each one of them is $1/N^n$. Conceptually, simple random sampling is the simplest of the probability sampling techniques. It requires a complete sampling frame, which may not be available or feasible to construct for large populations. Even if a complete frame is available, more efficient approaches may be possible if other useful information is available about the units in the population.

Simple random sampling is free of classification error, and it requires minimum advance knowledge of the population. It best suits situations where the population is fairly homogeneous and not much information is available about the population. If these conditions are not true, some other types of sampling techniques may be a better choice. Lottery method and computer generated random numbers are used to select a random sample in simple random sampling:

i) Lottery method: This is a very common method of taking a random sample under this method; we label each member of the population by identifiable ticket or pieces of papers.

Tickets must be of identical size, color and shape. They are placed in the container and well mixed before each draw and then draws may be continued until a sample of the required size is selected. This shows that selection of items depends entirely on chance.

Example 7.4: If we want to take a sample of 25 persons out of a population of 150, the procedure is to write the names of all the 150 persons on separate slips of papers, fold these slips, mix them thoroughly and then make a blindfold selection of 25 slips without replacement.

ii) Table of random numbers

This is an alternative method of selecting a simple random sample. It is constructed from the digits 0, 1, 2, ..., 9. There are several tables available in standard books of Statistics.

Suppose we want to select a sample of size n , then

Make a list of population to be sampled;

Give a distinct code number to each unit of the population;

Choose the direction of selection randomly;

Take n units whose code numbers coincide with the random numbers as numbers of the sample

By omitting those random numbers which do not exist on the list and repeated numbers if an element is not appear more than once in a sample.

Table of Random Numbers

Column								
Row	1	2	3	4	5	6	7	8
1	57172	42088	70098	17333	26902	29959	43909	49607
2	33883	87680	24923	15659	09839	45817	89405	70743
3	77950	15344	35609	87119	15859	74577	42791	75889
4	11607	26596	16796	24498	17009	67119	60557	49521
5	56149	55678	38169	47228	49931	94303	67448	31286
6	80719	65101	77729	83949	83358	75230	56624	27549
7	93809	19505	82000	79068	45552	86776	48980	56684

8	40950	86216	48161	17646	24164	35513	94057	51834
9	12182	59744	83710	41125	14291	74773	66391	50031
10	13382	48076	73151	48724	35670	38453	63154	58116
11	38629	94576	48859	75654	17152	66516	78796	73099
12	60728	52063	12431	23898	23683	10853	04038	75246
13	01881	99056	46747	08846	01331	88163	74462	14551
14	23094	08831	24387	23917	07421	97869	88092	72201

Example 7.5: Suppose that $N=40$ and we want to select $n=10$ without replacement, starting with the 3rd row and 2nd column by reading vertically using the above random table, we get

Solution: starting with the 3rd row and 2nd column by reading vertically we will get:

15, 26, 19, 08, 24, 35, 16, 38, 12 and 17.

b. Stratified random sampling

In stratified sampling, the population of N units is sub-divided into k sub-populations, called strata, so that the units in each stratum are as homogeneous as possible and the means of the different strata areas different as possible. These sub-populations should be non-overlapping so that they comprise the whole population such that $N_1 + N_2 + \dots + N_k = N$, where N_i represent the population size in the i^{th} strata. Then a sample is drawn from each stratum independently, the sample size within the i^{th} stratum being $n_i (i=1,2,\dots,k)$ such that $n_1 + n_2 + \dots + n_k = n$. The procedure of taking samples in this way is called stratified sampling. If the sample is taken randomly from each stratum, the procedure is known as stratified random sampling.

Remarks:

In stratified random sampling, the following two points are equally important to ensure accuracy. Proper stratification of the population into various strata, and

a suitable sample size from each stratum.

For example a population can be stratified based on the following variables:

- ✓ Sex (male, female)
- ✓ Age (under 18, 18 to 28, 29 to 39);
- ✓ Occupation (professional, other).
- ✓ Geographical classifications
- ✓ Administrative regions,etc.

c. Cluster Sampling:

The population is divided in to non-overlapping groups called clusters. A simple random sample of groups or cluster of elements is chosen and all the sampling units in the selected clusters will be surveyed in the case of single stage cluster sampling. Clusters are formed in a way that elements within a cluster are heterogeneous, i.e. observations in each cluster should be more or less dissimilar. Cluster sampling is useful when it is difficult or costly to generate a simple random sample. For example, to estimate the average annual household income in a large city we use cluster sampling, because to use simple random sampling we need a complete list of households in the city as sampling frame. To use stratified random sampling, we would again need the list of households. A less expensive way is to let each block within the city represent a cluster. A sample of clusters could then be randomly selected, and every household within these clusters could be interviewed to find the average annual household income.

d. Systematic Sampling:

Systematic sampling is the selection of every k^{th} element from a sampling frame, where k , the sampling interval and $k = \text{population size} / \text{sample size} = \frac{N}{n}$. Using this procedure each element in the population has a known and equal probability of selection. This makes systematic sampling functionally similar to simple random sampling. It is however, much more efficient and much less expensive to do. Like simple random sampling a complete list of all elements within the population (sampling frame) is required. The procedure starts in determining the first element to be included in the sample. It is however, much more efficient and much less expensive to do. Suppose that we have a complete and up-to-date list of the N units in the population numbered

from 1 to N in some order. To select a sample of size n, if N is an integral multiple of n, $N = kn$ for some integer k, $k = \frac{\text{population size}}{\text{sample size}} = \frac{N}{n}$.

The procedure starts in determining the first element to be included in the sample, select a unit i randomly from the first group, $i \leq k$ as the first element. The second unit will be $(i+k)^{\text{th}}$ element from the frame. Totality we have a sample of size n from the population of size N, i^{th} , $(i+k)^{\text{th}}$, $(i+2k)^{\text{th}}$, ... $(i+(n-1)k)^{\text{th}}$ element of the population are taken as a sample.

Example 7.6: Suppose that $N = 20$ and we want to select a sample of size 4, so that $k = N/n = 20/4 = 5$. The first element in the sample is selected from the first 5 units randomly, say 3^{rd} , which is the random start. Then, every 5^{th} unit is selected, and the sample contains the 3^{rd} , 8^{th} , 13^{th} and 18^{th} units of the population.

8.3.2 Non-Random Sampling or non-probability sampling

It is a sampling technique in which the choice of individuals for a sample depends on the basis of convenience, personal choice or interest.

Types of non-random sampling are:

1. Judgment sampling.
2. Convenience sampling
3. Quota Sampling.

1. Judgment Sampling

In this case, the person taking the sample has direct or indirect control over which items are selected for the sample. This method is mainly used for opinion surveys but is not recommended for general use, as it bias of the sampler.

2. Convenience Sampling

In this method, the decision maker selects a sample from the population in a manner that is relatively easy and convenient.

3. Quota Sampling

This is a type of judgment sampling and may be the most commonly used one in the non-probability category. In a quota sample, quotas are set up according to some specified characteristics such as income groups, age groups, political or religious groups, etc. Within the quota, the selection of sampling units depends up on personal judgment.

8.4 Sampling Distribution of the sample mean

Consider all possible samples of size n that can be drawn from a given population (either with or without replacement). For each sample, we can compute a statistic (such as the mean & the standard deviation) that will vary from sample to sample. In this manner we obtain a distribution of the statistic that is called its sampling distribution.

Steps for the construction of Sampling Distribution of the mean

1. From a finite population of size N , randomly draw all possible samples of size n . There are N^n possible samples if sampling is with replacement and there are ${}_N C_n$ possible samples if sampling is without replacement.
2. Calculate the mean for each sample.
3. Summarize the mean obtained in step 2 in terms of frequency distribution

Example 7.7: Suppose we have a population of size 5, consisting of the age of five children 3, 5, 7, 9, and 11. Population mean is 7 and population variance is 8. (Consider sampling without replacement).

Take samples of size 2 and construct sampling distribution of the sample mean.

Solution:

Step 1: $N=5$, $n=2$ we have ${}_5 C_2 = 10$, possible samples.

(3,5), (3,7), (3,9), (3,11), (5,7), (5,9), (5,11), (7,9), (7,11) and (9, 11)

Step 2: Calculate the sample mean for each sample:

Means = 4, 5, 6, 7, 6, 7, 8, 8, 9, 10 respectively.

Step 3: Summarize the mean obtained in step 2 in terms of frequency distribution.

$\bar{x}_i$	4	5	6	7	8	9	10	Total
f_i	1	1	2	2	2	1	1	10
$\bar{x}_i f_i$	4	5	12	14	16	9	10	70
$f(\bar{x}_i - 7)^2$	9	4	2	0	2	4	9	30

Mean of sample means, $E(\bar{X}) = \frac{\sum \bar{x}_i f_i}{\sum f_i} = 70/10 = 7$

Variance of sample means, $\text{var}(\bar{X}) = \frac{\sum_i^k (\bar{x}_i - E(\bar{X}))^2}{k} = 30/10 = 3$

$$V(\bar{x}) = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right) = \frac{8}{2} \left(\frac{5-2}{5-1} \right) = 3$$

Example 7.8

Three students have taken a class test which is marked out of 10. We want to estimate the mean mark using the sample mean as the estimate of the population mean. We take a sample of size 2 in two cases and suppose the marks of the three students are 1, 2 and 6.

The population mean μ is $(1+2+6)/3 = 3$

The population variance $\sigma^2 = \frac{\sum (x_i - \mu)^2}{N} = 14/3$

Sampling without replacement

In this type of sampling an observation is included in the sample only once and is selected randomly without any preference or conscious effort.

If sampling is without replacement we can take ${}_3C_2 = 3$ possible samples; the possibilities are given below.

Possible sample	(1,2)	(1,6)	(2,6)
Sample mean	1.5	3.5	4

The sample mean is a random variable, and we see that it can take three possible values. We can now write down its probability distribution as follows

$\bar{x}_i$	1.5	3.5	4	Total
$P(\bar{X} = \bar{x}_i)$	1/3	1/3	1/3	1
$(\bar{x}_i - 3)^2$	2.25	0.25	1	3.5

i) Mean of sample means $E(\bar{X}) = \sum \bar{x}_i p(\bar{X} = \bar{x}_i) = 1.5(1/3) + 3.5(1/3) + 4(1/3) = 3 = \text{population mean}$. i.e., Mean of sample means $E(\bar{X}) = \text{population mean}$

ii) Variance of sample means, $\text{var}(\bar{X}) = \frac{\sum_i^k (\bar{x}_i - E(\bar{X}))^2}{k} = 3.5/3 = 1.17$, where k is number of sample mean.

When sampling is without replacement, $V(\bar{x}) = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right) = \frac{14/3}{2} \left(\frac{3-2}{3-1} \right) = \frac{14}{12} = 1.17$.

Sampling with replacement

In this type of sampling an observation has a chance to be selected at each draw.

Suppose that we take the sample with replacement, there are $3^2 = 9$ possible samples.

Sample	(1,1)	(1,2)	(1,6)	(2,1)	(2,2)	(2,6)	(6,1)	(6,2)	(6,6)
--------	-------	-------	-------	-------	-------	-------	-------	-------	-------

Sample mean	1	1.5	3.5	1.5	2	4	3.5	4	6
-------------	---	-----	-----	-----	---	---	-----	---	---

The sample mean is a random variable & its probability distribution is:

$\bar{x}_i$	1	1.5	2	3.5	4	6	Total
$P(\bar{X} = \bar{x}_i)$	1/9	2/9	1/9	2/9	2/9	1/9	1
$\bar{x}_i P(\bar{X} = \bar{x}_i)$	1/9	1/3	2/9	7/9	8/9	6/9	3
$fi(\bar{x}_i - 3)^2$	4	4.5	1	0.50	2	9	21

i) Mean of sample means $E(\bar{X}) = \sum \bar{x}_i p(\bar{X} = \bar{x}_i) = 1(1/9) + 1.5(2/9) + 2(1/9) + 3.5(2/9) + 4(2/9) + 6(1/9) = 3$.

Mean of sample means, $E(\bar{X}) = \text{population mean}$.

ii) Variance of sample means $\text{var}(\bar{X}) = \frac{\sum_i^k (\bar{x}_i - E(\bar{X}))^2}{k} = 21/9 = 2.33$

Where k is number of sample means

$$V(\bar{X}) = \sigma_x^2 = \frac{\sigma^2}{n} = \frac{14/3}{2} = 14/6 = 2.33$$

In which if sampling with replacement, $V(\bar{X}) = \sigma_x^2 = \frac{\sigma^2}{n} = \frac{14/3}{2} = 14/6 = 2.33$.

In each case the expected value of the sample mean equals the population mean. This explains why the sample mean is a good estimate of the population mean. If we use the sample mean as an estimate of the population mean we will sometimes overestimate it, and sometimes underestimate it, but “on average” we will be accurate.

The example above illustrates an important result:

Remark:

1. Mean of sample means = $E(\bar{X}) = \frac{\sum \bar{x}_i f_i}{\sum f_i} = \sum \bar{x}_i p (\bar{X} = \bar{x}_i) = \text{population mean.}$

2. Variance of sample means, $V(\bar{X}) = \sigma_x^2 = \frac{\sigma^2}{n}$ (if sampling is with replacement).

3) Variance of sample means $V(\bar{x}) = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right)$, (if sampling is without replacement).

The quantity $\left(\frac{N-n}{N-1} \right)$ is finite population correction (fpc), and if $n/N < 0.05$, fpc is ignored.

Note: the square root the Variance of sample means is known as standard error.

The distribution of sample means depends on distribution of the population, sample size and whether population variance is known or unknown. A sample may be from a normally distributed population or from a non-normally distributed population, from a population with variance is known or unknown and the sample size may be large or small.

Case-I: If sampling is from a normally distributed population with known variance: When sampling is from a normally distributed population with known variance, the distribution of sample means, $\bar{X}$, is normal whatever the sample size.

Example 7.9: If an IQ of a class of students is normally distributed with mean 100 and variance 9

What is the probability that the average IQ of a class of 25 students exceeds 106?

Solution

Let X is IQ of students with mean 100 and variance 9.

A sample of size 25 is taken, the sample mean is a random variable ($\bar{X}$),

$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \Rightarrow \bar{X} \sim N\left(100, \frac{9}{25}\right)$, since the population is normally distributed,

$$P\left[\bar{X} \geq 106\right] = P\left[Z \geq \frac{106-100}{3}\right] = P[Z \geq 2] = 1 - 0.9772 = 0.0228$$

So, approximately 2% of average IQs of classes of 25 students will exceed 106.

Case-II: When sampling from a non-normal population and when the sample size is large

If sampling is from a non-normal population and when the sample size is large the distribution of $\bar{X}$ depends on Central Limit Theorem.

The Central Limit Theorem

If $X_1, X_2, \dots, X_n$ is a random sample from a population with mean μ and variance σ^2 , then as n goes to infinity the distribution of the sample mean, $\bar{X}$, approximates normal distribution with mean μ and variance $\frac{\sigma^2}{n}$. In short as n gets large number, $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$.

We can standardize this to get $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$ (approximately as n gets large). When

population variance is unknown $Z = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim N(0, 1)$ (approximately as n gets large).

Example 7.10: The mean weight of 500 male students at a certain university is 151 pounds (lb) and the standard deviation is 15 lb. By assuming that the weights are normally distributed and supposing that a sample of 64 students is taken, what is the probability that the weight in the sample is more than 154.75 lb?

Solution

As we have taken a large ($n=64$) sample we can use the Central Limit Theorem. This says that the mean weight of the sample can be approximated by a normal random variable with a mean of 151 and a variance of 225. If we let $\bar{X}$ be the mean weight of the students, it is required to find

$$P(\bar{X} > 154.75) =$$

$$\bar{X} \sim N\left(151, \frac{225}{64}\right) \Rightarrow P(\bar{X} > 154.75) = P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{154.75 - 151}{15/8}\right) = P(Z > 2.00) = 0.5 - 0.4772 = 0.0228$$

Example 7.11: Suppose that 150 customers enter a supermarket on a given day. Each customer spends a random amount. All they knew about the distribution of these expenditures that its mean is 7.50 birr and its standard deviation is 3.40 birr. What is the probability that a person, on average, spent more than 8.00 birr during the day?

Solution

We have $n = 150$ which is large enough to use the Central Limit Theorem. Mean = 7.50 and standard deviation = 3.40.

Let $\bar{X}$ be the mean amount of an individual's expenditure during the day. $\bar{X} \sim N(7.50, 0.077)$

Let $\bar{X}$ the average amount of an individual's expenditure during the day, it is required to find $P(\bar{X} > 8)$

$$P(\bar{X} > 8.00) = P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{8.00 - \mu}{\sigma/\sqrt{n}}\right) = P\left(Z > \frac{8.00 - 7.5}{3.4/\sqrt{150}}\right) = P(Z > 1.80) = 0.5 - P(0 < Z < 1.80)$$

$$= 0.5 - 0.4641 = 0.0359$$

This means there is only 0.0359 probabilities that a person will spent larger than 8.00 birr on average.

Case-III: When sampling is from normally distributed population with unknown population variance,

a) If the sample size is large, $Z = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim N(0, 1)$, where S is an estimate of σ .

b) If the sample size is small ($n < 30$), $t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{(n-1)}$. t has t-distribution with (n-1) degree of freedom, where S is an estimate of σ .

8.5 7.5 Sampling Distribution of the sample Proportion

In situations where it is not possible to measure the characteristic under study, but is possible to classify the whole population in various categories with respect to the attributes they possess, consideration is usually given to estimating the population elements that belong to a defined

category of class. Suppose that we have two complementary and mutually exclusive class, C and C' such that every unit in the population falls into either of them.

In order to know how many of the units fall in class C, we define a counting variable as

$$X_i = \begin{cases} 1, & \text{if the unit fall in class C} \\ 0, & \text{otherwise} \end{cases}$$

If the number of units falling in C is denoted by A for the population and by a for the sample, then

$$\sum_i^N X_i = A \text{ and hence the population proportion denoted by } P \text{ is given by } P = A/N$$

Given a simple random sample of n units, the sample proportion denoted by $p = \frac{a}{n} = \frac{\sum_i^n X_i}{n}$ from the formula, we see that $\bar{X}$ and p are essentially identical. In fact p is special case of $\bar{X}$, the case where possible values of X_i are only 0 and 1. Consequently p possesses all properties of $\bar{X}$.

p is an estimate of P, with variance

$$\text{var}(p) = \text{var}(\bar{X}) = \frac{\sigma^2}{n} \left(\frac{N-n}{N-1} \right) \text{ where } \sigma^2 = \frac{\sum_i^N (X_i - \mu)^2}{N} = PQ$$

$$\text{var}(P) = \frac{PQ}{n} \left(\frac{N-n}{N-1} \right)$$

Where $Q=1-P$ is proportion of units falling in class C'

$$\text{var}(P) = \frac{PQ}{n} \left(\frac{N-n}{N-1} \right) \text{ is estimated by using sample values as}$$

$$\text{var}(\hat{p}) = \frac{pq}{n-1} \left(\frac{N-n}{N} \right) = \frac{pq}{n-1} (1-f)$$

Where sampling fraction, $f = n/N$.

This expression is obtained by replacing σ^2 by its estimator $S^2 = \frac{npq}{n-1}$.

The sampling fraction can be ignored, when N is large relative to sample size n, $n/N < 0.05$.

$$\text{var}(\hat{p}) = \frac{pq}{n-1} \text{ and the standard error of } p \text{ is } \sqrt{\frac{pq}{n-1}}.$$

Sample proportion p is normally distributed with mean P and variance $\text{var}(p) = \frac{PQ}{n} \left(\frac{N-n}{N-1} \right)$

Example 7.12: In a simple random sample of size 100, from a population of size 500, there are 37 employed persons in the sample.

- a) Estimate proportion of employed persons in the population.
- b) Calculate the standard error of p.

Solution:

a) Population proportion P is estimated by $p = a/n = 37/100 = 0.37$. 37% of the population is employed.

b) Standard error of p is
$$\sqrt{\frac{pq(1-f)}{n-1}} = \sqrt{\frac{(0.37)(0.63)(1-0.2)}{99}} = 0.0434.$$

Exercise- 7

1. What is the difference between a statistic and a parameter?
2. What is a sampling frame?
3. An electrical firm manufactures light bulbs that have a length of life which are normally distributed with population mean 800 hrs and standard deviation 40 hrs. Find the probability that a bulb burns:
 - a) Between 778 and 834 hrs
 - b) Greater than 834 hrs?
4. The amount of sulphur in a daily emission from a factory has a normal distribution with mean of 134 pounds and a standard deviation of 22 pounds. For a day selected randomly, find the probability that the mean amount of sulphur emission will be less than 130 pounds.

Chapter-9

9 Estimation and Hypothesis Testing

The process of inferring information about a population from a sample is known as statistical inference. This chapter has two major parts. The first part is statistical estimation discusses the method of estimating a population parameter by using statistic, point estimation. It also explains the concepts of confidence interval. The second part is hypothesis testing describes the different techniques of testing a given tentative assumptions by applying an appropriate test statistic.

9.1 Statistical Estimation

It is the procedure of using a sample statistic to estimate a population parameter. This is one way of making inference about the population parameter where the investigator does not have any prior notion about values or characteristics of the population parameter. A statistic used to estimate a parameter is called an estimator and the value taken by the estimator is called *an estimate*. Statistical estimation is divided into two main categories: Point Estimation and Interval Estimation.

Point Estimation

When we use a single value of a statistic to estimate the corresponding parameter of a population, it is called point estimation. It is a common way of estimating a parameter, where a random sample of n observations is selected from a population and the statistic is calculated.

Examples:

- ✓ A sample mean is an estimate for population mean μ . That is, $\bar{X}$ is an estimator for population mean μ .
- ✓ A sample variance is an estimate for population variance. That is, S^2 is an estimator for population variance σ^2 .
- ✓ A sample proportion estimate for population proportion

Properties of best estimator

The following are some qualities of an estimator.

- ❖ It should be unbiased.
- ❖ It should be consistent.

- ❖ It should be relatively efficient.
- ❖ To explain these properties let $\hat{\theta}$ be an estimator of θ .

Unbiased Estimator: An estimator whose expected value is the value of the parameter being estimated. i.e., $E(\hat{\theta}) = \theta$.

Consistent Estimator: An estimator which gets closer to the value of the parameter as the sample size increases. i.e., $\hat{\theta}$ gets closer to θ as the sample size increases.

Relatively Efficient Estimator: The estimator for a parameter with the smallest variance.

This actually compares two or more estimators for one parameter.

Interval estimation

It is unlikely that any particular estimate will be exactly equal to the population mean, surely an estimate can be greater than or less than the parameter. That is, it is not always possible to estimate population parameter without any error so allowance is needed for such error. We take interval, ranges of values about an estimate in which the parameter may lie. This procedure is Interval estimation. It is the procedure that results in the interval of values of a parameter. Interval estimates indicate the precision or accuracy of an estimate and are, therefore, preferable to point estimates. It deals with identifying the upper and lower limits of a parameter. Confidence interval for the parameter is:

$$\text{Estimate} \pm \text{critical value} \times \text{Standard error of the estimator}$$

Example 8.1: Confidence interval for the population mean is: $\bar{X} \pm \text{Critical value} \times \text{Standard error of } (\bar{X})$

9.1.1 Confidence interval Estimation for population means

Although $\bar{X}$ possesses nearly all the qualities of a good estimator, because of sampling error, we know that it's not likely that our sample statistic will be equal to the population parameter, but instead will fall into an interval of values. We will have to be satisfied knowing that the statistic is "close to" the parameter. That leads to the obvious question, what is "close"?

We can phrase the latter question differently: How confident can we be that the value of the statistic falls within a certain "distance" of the parameter? Or, what is the probability that the

parameter's value is within a certain range of the statistic's value? This range is the *confidence interval*.

The *confidence level* is the *probability* that the value of the parameter falls within the range specified by the confidence interval surrounding the statistic. There are different cases to be considered to construct confidence intervals.

Case-I: Population variance (σ^2) is known and parent population is normal.

The sampling distribution of the sample mean is normal with mean μ and variance $\frac{\sigma^2}{n}$, that is,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

We can standardize this to get $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$.

From the standard normal distribution, we have

$$P(-Z_{\alpha/2} < Z < Z_{\alpha/2}) = 1 - \alpha$$

Where α is risk probability and $1 - \alpha$ confidence level. The confidence level is the probability that the value of the parameter falls within the range specified by the confidence interval surrounding the statistic. $\sigma/\sqrt{n}$ is the standard error of the statistic. Standard error is the square root of variance where $\text{Var}(\bar{X}) = \sigma^2/n$.

Using the standardized form of the sampling distribution of the sample mean in the above probability statement, we get the limits of the confidence interval as follows:

$$P\left(-Z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < Z_{\alpha/2}\right) = 1 - \alpha$$

$$P\left(-Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

$$P\left(-Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} - \bar{X} < -\mu < Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} - \bar{X}\right) = 1 - \alpha$$

$$P\left(\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

The last statement clearly shows that, there is a $(1 - \alpha)$ 100% confidence interval for population mean (μ) to lie in the interval

$$\left(\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right)$$

This interval is known as a $(1 - \alpha)$ 100% confidence interval for population mean (μ).

Here are the Z values corresponding to the most commonly used confidence levels.

$(1 - \alpha)$ 100%	α	$\alpha/2$	$Z_{\alpha/2}$
90	0.10	0.05	1.645
95	0.05	0.025	1.96
99	0.01	0.005	2.58

Example 8.2: The weights of full boxes of a certain kind of cereal are normally distributed with a standard deviation of 0.27 ounce. If a sample of 15 randomly selected boxes produced a mean weight of 9.87 ounce, find:

The 95% confidence interval for the true mean weight of boxes of this cereal,

The 99% confidence interval for the true mean weight of boxes of this cereal,

What effect does the increase in the level of confidence have on the width of the interval?

Solution:

Given $1 - \alpha = 0.95$, so that $\alpha/2 = 0.025$, $n = 15$, $\sigma = 0.27 \text{ ounce}$, $\bar{x} = 9.87 \text{ ounce}$. The 95% C.I. is $P(-Z_{0.025} < Z < Z_{0.025}) = 0.95$ and $\pm Z_{\alpha/2} = \pm Z_{0.025} = \pm 1.96$ ounce

Where $Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$.

Substituting these values in $\bar{x} - Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$, the resulting confidence interval is (9.73, 10.01).

Similarly the 99% C.I. is (9.69, 10.05).

The increase in the confidence level widens the length of the confidence interval.

Case-II: When sampling from a non-normal population and when the sample size is large the distribution of $\bar{X}$ depends on Central Limit Theorem (with known and unknown population variance).

Recall the *Central Limit Theorem*, which applies to the sampling distribution of the mean of a sample. Consider samples of size n drawn from a population, whose mean is μ and standard deviation is σ . The population can have any frequency distribution. The sampling distribution of $\bar{X}$ will have a mean μ and standard deviation is $\frac{\sigma}{\sqrt{n}}$. The sampling distribution of $\bar{X}$ is normal with a mean μ and variance $\frac{\sigma^2}{n}$ as n gets large. That is $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ (as n gets large). We can standardize this to get $Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0,1)$ or $Z = \frac{\bar{X} - \mu}{S / \sqrt{n}} \sim N(0,1)$ when σ is unknown.

A $(1-\alpha)$ 100% confidence interval for population mean (μ) is

$(\bar{X} - Z_{\alpha/2} \sigma / \sqrt{n}, \bar{X} + Z_{\alpha/2} \sigma / \sqrt{n})$ if σ^2 is known and

$(\bar{X} - Z_{\alpha/2} S / \sqrt{n}, \bar{X} + Z_{\alpha/2} S / \sqrt{n})$ if σ^2 is unknown.

Example 8.3: An economist wants to estimate the average amount in checking accounts at banks in given region. A random sample of 100 accounts gives $\bar{X} = \$357.60$ and $S = \$140.00$. Give a

95% confidence interval for μ , the average amount in any checking account at a bank in the given region.

Solution

Given: $n = 100, \bar{X} = \$357.60, S = \140.00 & $\alpha = 0.05$

A 95% confidence interval for population mean (μ) is

$$(\bar{X} - Z_{\alpha/2} S/\sqrt{n}, \bar{X} + Z_{\alpha/2} S/\sqrt{n}) \dots \text{since } n \text{ is large and } \sigma^2 \text{ is unknown}$$

$$=(357.60 - 1.96(140.00/\sqrt{100}), 357.60 + 1.96(140.00/\sqrt{100}))$$

$$= (330.16, 385.04).$$

Case-III: When sampling is from normally distributed population with unknown population variance and when the sample size is small ($n < 30$).

When population variance σ^2 is unknown, we estimate it by sample variance. The standardized distributions of the sample mean, $t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$ is t-distribution with $(n-1)$ degrees of freedom. From this distribution, $(1-\alpha)$ 100% confidence interval for population mean is

$$(\bar{X} - t_{\alpha/2(n-1)} \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2(n-1)} \frac{S}{\sqrt{n}}).$$

Example 8.4: From a normal sample of size 25 a mean of 32 was found .Given that the standard deviation is 4.2. Find

- A 95% confidence interval for the population mean.
- A 99% confidence interval for the population mean.

Solution:

Given: $n = 25, \bar{X} = 32, S = 4.2, 1-\alpha = 0.95 \Rightarrow \alpha = 0.05, \frac{\alpha}{2} = 0.025$

$$\Rightarrow t_{\alpha/2, 24} = 2.064 \text{ from table.}$$

$\Rightarrow$ The required interval will be $\left(\bar{X} - t_{\alpha/2(n-1)} \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2(n-1)} \frac{S}{\sqrt{n}} \right)$

$$= 32 \pm 2.064 \frac{4.2}{\sqrt{25}} = 32 \pm 1.73 = (30.27, 33.73)$$

Given: $n = 25$, $\bar{X} = 32$, $S = 4.2$, $1 - \alpha = 0.99 \Rightarrow \alpha = 0.01$, $\frac{\alpha}{2} = 0.005$

$$\Rightarrow t_{\alpha/2, 24} = 2.797 \text{ from table.}$$

$\Rightarrow$ The required interval will be $(\bar{X} - t_{\alpha/2(n-1)} \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2(n-1)} \frac{S}{\sqrt{n}})$

$$= 32 \pm 2.797 \times \frac{4.2}{\sqrt{25}}$$

$$= 32 \pm 2.35$$

$$= (29.65, 34.35)$$

9.1.2 Sample size determination in estimation of population mean

In the process of estimating population mean μ using the sample mean with absolute margin of error (d) and risk probability α , the sample size is given by:

$$n = \left[\frac{Z_{\alpha/2} \sigma}{d} \right]^2 \text{ where } |\bar{X} - \mu| = d$$

Example 8.5: To determine the average amount of time students take to get from one class to the next, how large a sample is needed with probability 0.95 that the error will be at most 0.25 minutes, if σ is known from past experience to be 1.50 minutes?

Solution: Using $Z_{0.025} = 1.96$, and replacing $E = 0.25$, and $\sigma = 1.50$ in the formula for n , we get $n = 138.30 \approx 139$ (always rounded to the next integer) is required for the estimate.

9.1.3 Confidence interval for population proportion

The confidence interval for the population proportion is performed in the same manner as the population mean. We have discussed that the sampling distribution of sample proportion is

normal .The sample estimate of population proportion P is sample proportion p and sample estimate of variance of sample proportion is $Var(\hat{p}) = \frac{pq}{n-1}$ for large sample $Var(\hat{p}) = \frac{pq}{n}$.

A $(1-\alpha)100\%$ confidence interval for proportion p is given by (for large n):

$$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$$

Example 8.6: The Human Resource director of a large organization wanted to know what proportion of all persons who had ever been interviewed for a job with his organization had been hired. He was willing to settle for 95% confidence interval. A random sample of 500 interview records revealed that 76 or 0.152 of the persons in the sample had been hired.

Solution:

Given: $\hat{p} = 0.152, \hat{q} = 1 - \hat{p} = 0.848, n = 500, \alpha = 0.05, Z_{0.025} = 1.96$

The 95% confidence interval for the population proportion is given by

$$\begin{aligned} \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} &= 0.152 \pm 1.96 \sqrt{\frac{0.152 \times 0.848}{500}} = 0.152 \pm 0.031 \\ &= \underline{(0.121, 0.183)} \end{aligned}$$

Hence the required proportion varies between 0.121 and 0.183.

9.2 Statistical Hypothesis testing

In section 8.1, we have studied how to make estimations of the mean using point and interval estimations. The other aspect of statistical inference is known as statistical test of hypothesis. The branch of statistics which helps us in arriving at the criterion for deciding about the characteristics of the population, a parameter, based on the information obtained from the sample data is known as testing of hypothesis. We shall use the theoretical results presented for the interval estimation, and hence, a test of hypothesis is highly connected with the theory of estimation we studied before.

In this section, basically we will deal with testing hypotheses about population mean and population proportion. While doing so, we shall define some important terminologies which we

may face and the errors we are committing in the process. We shall employ the standard normal distribution (or Z-test) and the t-distribution (or t-test), depending upon the nature of the population sampled and the sample.

9.2.1 Hypothesis testing for population mean

9.2.1.1 Some terms in tests of hypothesis

Statistical hypothesis is defined as a statement (or an assertion) about the parameter of a population or its distribution that may be proved or disproved. Its plausibility is to be evaluated on the basis of information obtained by taking sample from the population.

Test statistic: is a statistic whose value serves to determine whether to reject or accept the hypothesis to be tested. It is a random variable.

A given statement concerning a parameter could be true or false. Hence we have two complementary hypotheses, namely, null hypothesis and alternative hypothesis.

a) Null hypothesis (H_0)

It is the hypothesis to be tested for possible rejection under the assumption that it is true and it is the hypothesis of equality or the hypothesis of no difference.

b) Alternative hypothesis (H_1)

It is hypothesis which is the complementary to the null hypothesis. It may be accepted if H_0 is rejected or be rejected if H_0 is accepted; It is the hypothesis of difference.

Statistical Test: is a test or procedure used to evaluate a statistical hypothesis for deciding whether to reject the hypothesis depending on sample data. The decisions we make are of two types: Either to reject H_0 and conclude that H_1 is accepted or retain H_0 and conclude that we have no enough evidence to reject H_0 .

Types of errors

Statistical test of hypothesis can lead to two kinds of errors. If the statistical test rejects H_0 when it is true, the error is type I error. If the test accepts H_0 when it is false, the error is a type II error.

The following table gives a summary of possible results of any hypothesis testing procedure:

Decision	H_0 is true	H_0 is false
Reject H_0	<i>Type I error</i>	Correct decision
Accept H_0	Correct decision	<i>Type II error</i>

Type I error is the error committed in rejecting the null hypothesis when it is true. Probability of committing *type I error* is sometimes called level of significance and denoted by α .

Type II error is the error committed in accepting the null hypothesis when it is false. Probability of committing *type II error* is denoted by β .

In both types of errors, a wrong decision has occurred. An ideal test procedure is one which is so planned as to safeguard against both these errors. However, in practical situations an attempt to minimize one of the errors maximizes the other. In view of this dilemma and the fact that wrong rejection of H_0 is a more serious error, we will hold α at a predetermined low level, such as 0.1, 0.05, or 0.01 when choosing a rejection region. The level of significance 5% ($\alpha = 0.05$) implies that in 5 samples out of 100 we are likely to reject a correct H_0 . In other words this implies that we are 95% confident that our decision to reject H_0 is correct.

General steps in hypothesis testing on population mean, μ

Step-1 The first step in hypothesis testing is to specify the null hypothesis (H_0) and the alternative hypothesis (H_1). Suppose the assumed or hypothesized value of μ is denoted by μ_0 , then one can formulate two sided and one sided hypothesis as follows:

1. $H_0: \mu = \mu_0$ versus $H_1: \mu \neq \mu_0$ (two sided test)
2. $H_0: \mu = \mu_0$ versus $H_1: \mu < \mu_0$ (one sided test)
3. $H_0: \mu = \mu_0$ versus $H_1: \mu > \mu_0$ (one sided test)

Step-2: Specify a significance level of α .

Step-3 We should identify the sampling distribution of the estimator and the test statistic.

Case-I: Population variance (σ^2) is known and parent population is normal.

The test statistic is $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$.

Case-II: When sampling from a non-normal population and when the sample size is large the distribution of X depends on *Central Limit Theorem* (with known and unknown variance).

a) The test statistic is: $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$ with known variance

b) The test statistic is: $Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim N(0, 1)$ with unknown variance.

Case-III: When sampling is from normally distributed population with unknown population variance.

When the sample size is large, $Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim N(0, 1)$.

When the sample size is small ($n < 30$), $t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{(n-1)}$.

Step-4. The value of the test statistic can be calculated as follows:

$Z_c = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$ with known variance.

$Z_c = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$ with unknown variance & large sample size.

$t_c = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$ with unknown variance and small sample size.

where $\bar{X}$ is the sample mean and μ_0 the parameter specified by the null hypothesis.

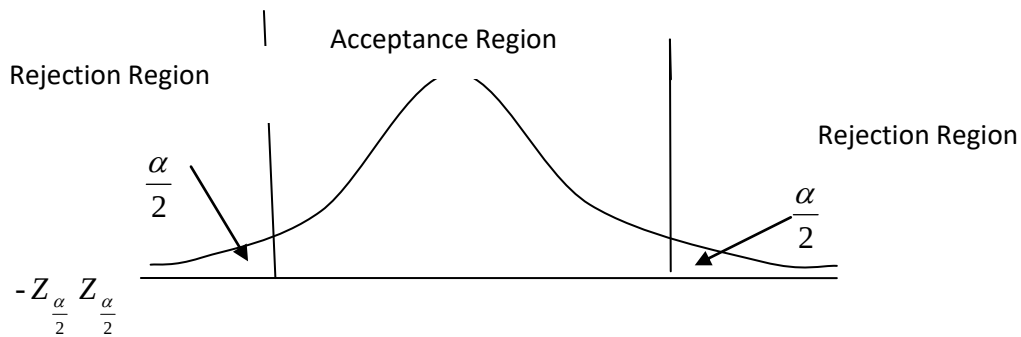
Step-5: Identify the critical (rejection) region or put the decision rule.

a) For two sided test $H_0: \mu = \mu_0$ versus $H_1: \mu \neq \mu_0$, reject H_0 if

$$Z_c > Z_{\alpha/2} \text{ or } Z_c < -Z_{\alpha/2}.$$

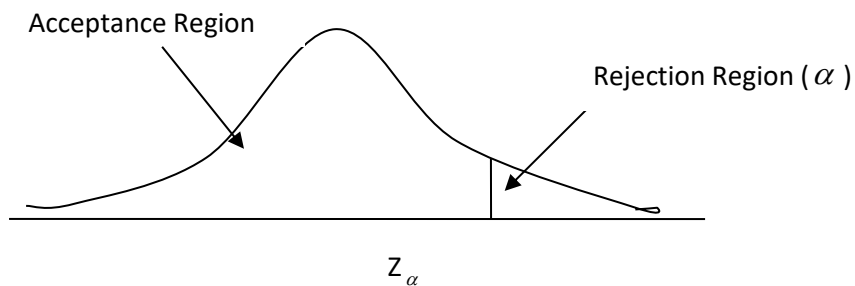
Note: Z_c refers to $Z_{\text{calculated}}$

Graphically, the rejection and acceptance regions are:



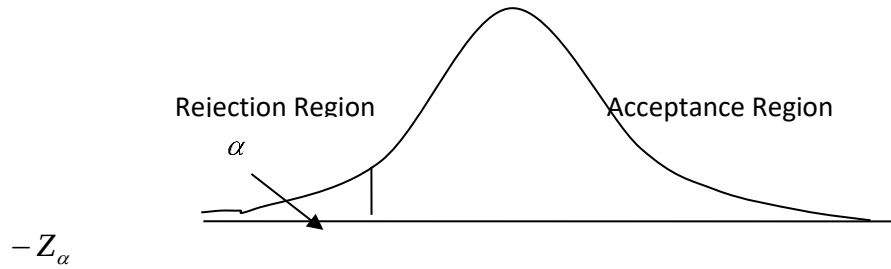
b) For one sided test (right sided test) $H_0: \mu = \mu_0$ versus $H_1: \mu > \mu_0$ reject H_0 if $Z_{\text{calculated}} > Z_{\alpha}$.

Graphically, the rejection and acceptance regions are



C) For one sided test (left sided test) $H_0: \mu = \mu_0$ versus $H_1: \mu < \mu_0$

reject H_0 if $Z_{\text{calculated}} < -Z_{\alpha}$. Graphically, the rejection and acceptance regions are



Step 6: Summarization the result and put the conclusion

Decision Table

To test $H_0 : \mu = \mu_0$ against the three alternatives, the rules are summarized as:

Alternative Hypothesis	Accept H_0 if	Reject H_0 if	Inconclusive if
$\mu \neq \mu_0$	$-Z_{\alpha/2} < Z_C < Z_{\alpha/2}$	$Z_C > Z_{\alpha/2}$ or $Z_C < -Z_{\alpha/2}$	$Z_C = Z_{\alpha/2}$ or $Z_C = -Z_{\alpha/2}$
$\mu > \mu_0$	$Z_C < Z_{\alpha}$	$Z_C > Z_{\alpha}$	$Z_C = Z_{\alpha}$
$\mu < \mu_0$	$Z_C > -Z_{\alpha}$	$Z_C < -Z_{\alpha}$	$Z_C = -Z_{\alpha}$

Example 8.7: Test at $\alpha = 0.05$ whether the mean of a random sample of size $n = 16$ is "significantly less than 10" if the distribution from which the sample was taken is normal, $\bar{x} = 8.4$ and $\sigma = 3.2$ (known).

Solution:

* $H_0 : \mu = 10$ versus $H_A : \mu < 10$, $\alpha = 0.05$

* $Z_{\alpha} = Z_{0.05} = 1.645$ (critical value)

* $Z_C = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{8.4 - 10}{3.2 / 4} = -2$ (calculated value)

* Since $Z_c = -2 < -Z_\alpha = -1.645$, the null hypothesis is rejected. That is, the population mean 8.4 is significantly less than 10 at 5% level of significance.

Example 8.8: Based upon a random sample of size 100 with an average of 3.4 minutes and a standard deviation of 2.8 minutes, is the claim that the average telephone call is 4 minutes true with a confidence of 95%?

Solution: Given: $n=100$, $\bar{x}=3.4\text{min}$, $s=2.8\text{ min}$, $\alpha=0.05$

To test: $H_0 : \mu = 4$
 $H_A : \mu \neq 4$

Since σ is unknown this should be a t-distribution; however, since $n=100$ is large the z-statistic is used.

$$Z_c = \frac{\bar{X} - \mu_0}{S / \sqrt{n}} = \frac{3.4 - 4}{2.8 / 10} = -2.14$$

From the standard normal table we have, $\pm Z_{\alpha/2} = \pm Z_{0.025} = \pm 1.96$

Since the calculated value is less than the tabulated value ($-2.14 < -1.96$), the null hypothesis will be rejected. Therefore average telephone call is significantly different from 4 minutes at $\alpha=0.05$.

Example 8.9: A sample of 16 students gave an average mark of 53.8 with a standard deviation of 5.2. Can we conclude that the population mean of marks is 50 at $\alpha=0.05$?

Solution: $H_0 : \mu = 50$ $H_A : \mu \neq 50$

$\alpha=0.05$ and hence $\alpha/2=0.025$

$$t_{\alpha/2, n-1} = t_{0.025, 15} = 2.131.$$

$$t_c = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} = \frac{53.8 - 50}{5.2 / \sqrt{16}} = \frac{3.8}{1.3} = 2.92.$$

Since $t_c = 2.92 > 2.131$, H_0 is rejected. Therefore the population mean mark is significantly different from 50 at $\alpha = 0.05$.

9.2.2 Hypothesis testing for population proportion

Hypothesis testing for population proportion P is carried out in the same way as hypothesis testing for population mean when large samples and normality assumptions are fulfilled.

The test statistic is:

$$Z = \frac{\hat{P} - p_o}{\sqrt{\frac{p_o q_o}{n}}} \sim N(0, 1) \text{ where } q_o = 1 - p_o$$

The decision rule is:

a) For two sided test $H_0: P = P_o$ versus $H_1: P \neq P_o$ reject H_0 if

$$Z_c > Z_{\alpha/2} \text{ or } Z_c < -Z_{\alpha/2}.$$

b) For one sided test $H_0: P = P_o$ versus $H_1: P > P_o$ reject H_0 if

$$Z_{\text{calculated}} > Z_{\alpha}$$

c) For one sided test $H_0: P = P_o$ versus $H_1: P < P_o$, reject H_0 if

$$Z_c < -Z_{\alpha}.$$

Example 8.10: A sales clerk in the departmental store claims that 60% of the shoppers entering the store leave without making a purchase. A random sample of 50 shoppers showed that 35 of them left without buying anything. Are these sample results consistent with the claim of the sales clerk? Use a level of significance of 0.05.

Solution: Given: $P_o = 0.6$, $q_o = 0.4$, $\hat{P} = \frac{35}{50} = 0.7$, $n = 50$, $\alpha = 0.05$, $Z_{0.05} = 1.645$

$H_0: P = 0.6$ Vs $H_1: P > 0.6$

Using the z statistic, we have

$$Z_c = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 q_0}{n}}} = \frac{0.7 - 0.6}{\sqrt{\frac{0.6 \times 0.4}{50}}} = 1.44$$

Since computed value of $Z_c = 1.44$ is less than the critical value of $Z_{0.05} = 1.645$, therefore, the null hypothesis cannot be rejected. Hence, based on this sample data we cannot reject the claim of the sales clerk.

9.3. Test of association

A Chi-Square (χ^2) is a probability distribution used to make statistical inferences about the association of categorical data (proportions) in which the numbers of categories are two or more.

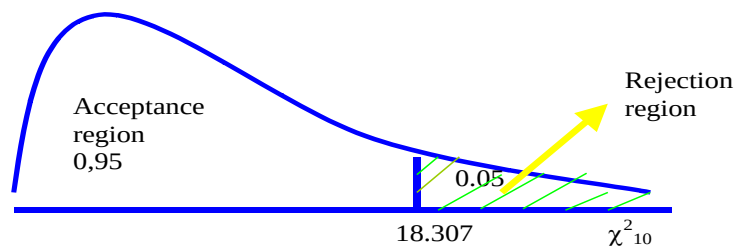
Widely used in the analysis of contingency tables. Chi-Square test allows us to test for association between two categorical variables.

H_0 : No association between the variables

H_A : There is association

Consequently a significant *p-value* implies association.

- Chi-Square test compares observed to expected counts (frequencies) under the assumption of no association (or H_0 is true)
- With this method, data are arranged in the form of a contingency table.
- Indexed by the degrees of freedom (n)
- Unlike z and t distributions, which are always symmetric about 0, the X^2 distribution only takes on positive values and is always skewed to the right.
- The skewness diminishes as n increases



Assumptions of the χ^2 - test

- No expected value in the table is <5 , and
- No observed frequency is zero
- No more than 20% of the **expected** frequencies should be <5 .
- Observation should be independent of each other
- There must be large sample (>40)

Expected Value: Is the product of the row total multiplied by the column total, divided by the grand total

$$E = \frac{\text{row total} \times \text{column total}}{\text{Overall total}}$$

The expected numbers must be computed for each cell.

Example 8.12: Observed Numbers and Response by Treatment

Treatment	Response		Total
	Yes	No	
A	37 (74%)	13 (26%)	50
B	17 (24.3%)	53 (75.7%)	70
Total	54	66	120

Expected Numbers

Treatment	Response		Total
	Yes	No	
A	0.45(50)=22.5	27.5	50
B	0.45(70)=31.5	38.5	70
Total	54 (45%)	66 (55%)	120

$$\begin{aligned}
 \chi^2 &= \sum_i \left[\frac{(O_i - E_i)^2}{E_i} \right] \\
 &= \frac{(37 - 22.5)^2}{22.5} + \frac{(13 - 27.5)^2}{27.5} + \frac{(17 - 31.5)^2}{31.5} + \frac{(53 - 38.5)^2}{38.5} \\
 &= \frac{210.25}{22.5} + \frac{210.25}{27.5} + \frac{210.25}{31.5} + \frac{210.25}{38.5}
 \end{aligned}$$

χ^2 statistic with 1 df = 29.1

Since χ^2 statistic = 29.1 exceeds the critical value of 3.84 for 1 df ($\alpha=0.05$), we reject the null hypothesis of no association.

Exercise - 8

1. From a normal population with the standard deviation is 4.2. A sample of size 25 is taken with mean of 32. Find 99% confidence interval for the population mean.
2. Out of a sample of 80 customers 60 of them reply they are satisfied with the service they received .Calculate a 95% confidence interval for the proportion of satisfied customers.
3. According to the norms established for a reading comprehension test, students should average 84. If 45 randomly selected students averaged 87.8 with standard deviation of 8.6, test the null hypothesis $\mu = 84$ against the alternative $\mu > 84$, at $\alpha = 0.01$.
4. In a study of phobia, a psychologist claims that 30% of all women are afraid of flying. If, in a random sample, 41 of 150 women are afraid of flying, test the null hypothesis $p = 0.30$ against $H_A : p \neq 0.30$, at $\alpha = 0.05$.

CHAPTER NINE

10. Simple Linear Regression and Correlation

10.1. Simple Linear Regression of Y on X

Under simple linear regression of Y on X, we have one independent variable which is influential usually denoted by X and one dependent variable influenced by the independent variable which we denote it by Y. For example in real world variables that may be related linearly are, production/yield (Y) and amount of rainfall(X), monthly income (Y) and level of education (X), ,where an increase in one variable is associated with an increase in the other variable. Similar examples can also be given on the negative relation between two variables; the increase in one is accompanied by a decrease in the other.

A simple linear regression model is given as

$$Y = \alpha + \beta X + \epsilon$$

Where α is intercept of the regression line. It gives the value of Y whenever X is zero. If the range of X does not include zero, α has no practical interpretation. β is the slope. It is a measure of the rate of change. It shows by how much Y changes for every unit change in X. The sign of β has also some significance; because it shows the direction of the relation between the two variables. A positive sign of β shows that the two variables are positively related and a negative sign of β shows that the two variables are negatively related.

The constants, α and β are parameters and are commonly referred to as regression coefficients.

- ϵ is a random error term. It is neither observable nor measurable. In real life problems, even though two variables are linearly related, their relationship is not fixed as

$$Y = \alpha + \beta X$$

This is because the dependent variable, Y is the effect of many independent variables in which X is one of them. Contribution of other independent variables not considered in the model may be minor. However, we cannot be certain that Y depends only on X. Thus the contribution of these

variables not included in the model and other factors such as measurement error is accommodated by ϵ .

Mean of the values of ϵ is zero. Some of its values are positive, that is when the actual value lies above the line $\hat{Y} = \hat{\alpha} + \hat{\beta}X_i$ and some are negative in case when the actual value of Y lies below the fitted regression line.

Assumptions:

1. The relationship between the dependent variable Y and independent variable X exist and is linear.
2. For every value of the independent variable X , there is an expected value of the dependent variable Y .
3. The dependent variable Y is a continuous random variable, whereas values of the independent variable X are fixed values.
4. The sampling error ϵ , associated with the expected value of the dependent variable Y is assumed to be an independent random variable distributed normally with mean 0 and constant variance σ^2 about the regression line.

To estimate this model we take a sample of n independent observations which give rise to n pairs (X_i, Y_i) and find best estimates of the parameters or best fitted line using least square method of estimation. A best fitting line is one for which the sum of squares of the errors, $\sum \epsilon_i^2$ is minimum.

In the principle of least square method, one would select α and β such that

$$\sum \epsilon_i^2 = \sum (Y_i - \hat{Y}_i)^2 \text{ is minimum where } \hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i$$

To minimize this function, first we take the partial derivatives of $\sum \epsilon_i^2$ with respect to $\hat{\alpha}$ and $\hat{\beta}$ respectively. Then the partial derivatives are equated to zero separately and result in the following normal equations respectively

$$\sum_{i=1}^n Y_i = n\hat{\alpha} + \hat{\beta} \sum_{i=1}^n X_i$$

$$\sum_{i=1}^n X_i Y_i = \hat{\alpha} \sum_{i=1}^n X_i + \hat{\beta} \sum_{i=1}^n X_i^2$$

Solving these normal equations simultaneously we can get the values of $\hat{\alpha}$ and $\hat{\beta}$ as follows.

$$\hat{\beta} = \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2} = \frac{n \sum x_i y_i - \left(\sum x_i \right) \left(\sum y_i \right)}{n \sum x_i^2 - \left(\sum x_i \right)^2} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} \text{ and } \hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X}$$

These estimates are denoted by $\hat{\alpha}$ and $\hat{\beta}$. The estimated (fitted) regression line is given by:

$$\hat{Y}_i = \hat{\alpha} + \hat{\beta} X_i$$

Before estimating the regression coefficients, it would be wise to plot the observed data on a graph known as a scatter diagram. Scatter diagram is a plot of all ordered pairs (x_i, y_i) on the coordinate plane which helps to observe relationship between two variables. This diagram gives

Student	1	2	3	4	5	6	7	8	9	10	Total
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a preliminary idea on the type of relationship the two variables have.

Regression analysis is useful in predicting the value of one variable from the given value of another variable, $\hat{Y}_i = \hat{\alpha} + \hat{\beta} X_i$.

Example 9.1: For the following example [the number of hours (X) a student spent studying and the marks (Y) each student received in an examination]:

Assuming simple linear relationship between X and Y,

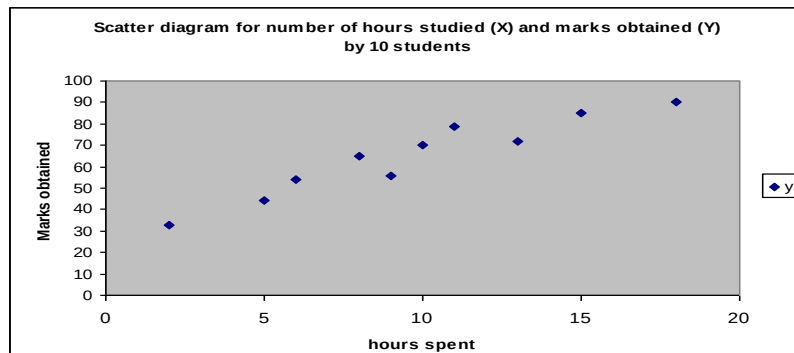
a/ Draw the scatter diagram;

b/ Find the estimated regression equation of Y on X;

c/ Give the predicted value of Y for X= 12

x	8	5	11	13	10	6	18	15	2	9	97
y	65	44	79	72	70	54	90	85	33	56	648
x ²	64	25	121	169	100	36	324	225	4	81	1149
xy	520	220	869	936	700	324	1620	1275	66	504	7034
y ²	4225	1936	6241	5184	4900	2916	8100	7225	1089	3136	44952

Solution: a) The scatter diagram is as follows:



b) And the necessary statistics are computed below:

$$\hat{\beta} = \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2} = \frac{7034 - (10)(9.7)(64.8)}{1149 - (10)(9.7)^2} = \frac{748.4}{208.1} = 3.596 \text{ and } \hat{\alpha} = 64.8 - 3.596(9.7) = 29.92.$$

Hence, the equation is $\hat{y} = 29.92 + 3.596x$.

c) When $X = 12$, $\hat{y}_{12} = 29.92 + 3.596(12) = 73.1$.

10.2. Simple Linear Correlation Analysis

Given the paired data $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ we may want to describe the type & strength of relationship between the independent variable X and the dependent variable Y. We can give these two by applying an index called simple correlation coefficient. The population correlation coefficient is represented by ρ and its estimator by r. The correlation coefficient r is also called Pearson's correlation coefficient since it was developed by Karl Pearson. The computational formula is:

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

Alternatively: The correlation coefficient is given by

$$r = \frac{\sum xy - n\bar{x}\bar{y}}{\sqrt{(\sum x^2 - n\bar{x}^2)(\sum y^2 - n\bar{y}^2)}}$$

The correlation coefficient, r is always lies between -1 and $+1$, inclusive.

- $r = -1$ implies perfect negative linear relationship between the two variables.
- $r = +1$ implies perfect positive linear relationship between the two variables.
- $r = 0$ implies there is no linear relationship between the two variables. But the two variables may have non-linear relationship between them.
- r approaches $+1$ indicates strong positive linear relationship between the two variables.
- r approaches -1 indicates strong negative linear relationship between the two variables.
- r approaches 0 indicates weak linear relationship between the two variables .

10.3. Coefficient of Determination (r^2)

The square of the correlation coefficient, r^2 , is called the coefficient of determination. It measures the variation in the dependent Y explained by the simple linear regression of Y on X .

$1 - r^2$ measures the proportion of variation in Y not explained by the simple linear regression of Y on X .

Example 9.2: If $r = 0.9$, then $r^2 = 0.81$ and $1 - r^2 = 0.19$. Approximately 81% of the variation in the dependent variable, Y , is explained by the simple linear regression of Y on X . The remaining, $1 - r^2$, 19 % of the variation in Y is unexplained by the simple linear regression of Y on X .

Example 9.3: The research director of the Saving and Loan Bank collected 25 observation of mortgage interest rates X and number of house sales Y at each interest rate. The director computed that,

$$\sum x_i = 125, \sum y_i = 100, \sum x_i y_i = 520, \sum x_i^2 = 650, \sum y_i^2 = 436$$

Compute and interpret

(i) Coefficient of correlation.

(ii) The coefficient of determination.

Solution: i) Coefficient of correlation.

$$r = \frac{\sum xy - n\bar{x}\bar{y}}{\sqrt{(\sum x^2 - n\bar{x}^2)(\sum y^2 - n\bar{y}^2)}} = \frac{520 - (25)(5)(4)}{\sqrt{(650 - 25(5)(5))(436 - (25)(4)(4))}} = 0.667$$

The two variables have positive linear relationship.

ii) Coefficient of determination, $r^2 = (0.667)^2 = 0.44$ this shows that 44% of the variation in the number of house sales is due to the variation in the interest rate.

10.4. Spearman's Rank Correlation Coefficient

The simple correlation coefficient (r) cannot be used when we are dealing with a qualitative data such as judgment about beauty, efficiency, honesty, etc. In such cases, the rank correlation coefficient is used to explain the correlation or if there is an agreement in ranking. It is denoted by r_s and is defined as follows:

Definition: The coefficient of rank correlation, r_s given by Spearman for n pairs, is

$$r_s = 1 - \frac{6\sum d^2}{n(n^2 - 1)}; \text{ Where } d \text{ is the difference between the rank of } x \text{ and the corresponding } y.$$

To calculate r_s , we first rank the x 's among themselves from least to best or from best to least; then we rank the y 's in the same way, find the sum of the squares of the differences, d , between the ranks of the x 's and the y 's. When there are ties in rank, we assign to each of the tied observations (having equal value) the mean of their ranks.

Example 9.4: Assume that ten girls in a beauty contest for Miss DebreMarkos were ranked by two judges as follows:

Girl Number	1	2	3	4	5	6	7	8	9	10
Judge A	4	8	6	7	1	3	2	5	10	9
Judge B	3	9	6	5	1	2	4	7	8	10

Calculate r_s and interpret it.

Solution: Since the ranks are given, we need to find only the difference in ranks for each girl and the square of these differences.

Girl Number	1	2	3	4	5	6	7	8	9	10	Total
D	1	-1	0	2	0	1	-2	-2	2	-1	0
d^2	1	1	0	4	0	1	4	4	4	1	20

For these $n = 10$ pairs, $\sum d^2 = 20$, and $r_s = 1 - \frac{6(20)}{10(100-1)} = 0.88$, which is positive and close to 1,

showing that there is a very good agreement (or concordance) between the two judges regarding the beauty of the girls.

N.B: $\sum d = 0$ provides a check in calculations.

Like the values of r , the values of r_s also lie between -1 and +1, inclusive, and the interpretations of its size and sign are analogous to those of r . $r_s = +1 \Rightarrow$ Perfect positive agreement, $r_s = -1 \Rightarrow$ complete disagreement where the two rankings go completely in opposite direction.

Exercise 9

1. What is scatter diagram? What is the advantage of scatter diagram?
2. What is the coefficient of determination
3. Based on the following data answer the question.

Sales	15	18	25	27	30	35
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Advertising expenditure	50	65	82	95	110	120
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- Decide which variable should be the independent variable and which should be the dependent variable.
 - Make a scatter plot of the data.
 - Does it appear from inspection that there is a relationship between the variables?
 - Calculate the least squares line. Put the equation in the form of: $\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i$
 - Find and interpret the correlation coefficient.
 - What is the slope of the least squares (best-fit) line? Interpret the slope.
4. Below are the planets distance from the sun and the time it takes for the planet to complete its orbit around the sun.

Planet	Distance from the sun (millions of miles)-x	Year to complete the orbit-y
Mercury	36	0.24
Venus	67	0.62
Earth	93	1.00
Mars	142	1.88
Jupiter	483	11.9
Saturn	887	29.5
Uranus	1785	84.0
Neptune	2797	165.0
Pluto	3675	248.0

- Make a scatter plot of the data. Does it appear from inspection that there is a linear relationship between the variables?
- Calculate the least squares line. Put the equation in the form of: $\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i$

c. Find and interpret the correlation coefficient.

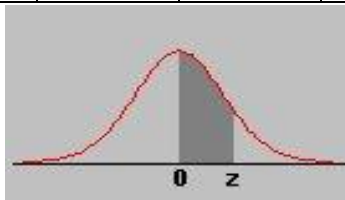
d. What is the slope of the least squares (best-fit) line? Interpret the slope.

e. find the estimated year to complete the orbit if the distance is 1000.

5. The number of cigarettes consumed (in billions)(say, x) and the number of cigarettes exported from the same country (say, y).

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
---	------	------	------	------	------	------	------	------	------	------

X	525	510	500	485	486	487	486
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y	164	179	206	196	220	231	244
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Compute and interpret

a) Calculate the least squares line. Put the equation in the form of: $\hat{Y}_i = \hat{\alpha} + \hat{\beta}X_i$

b) The Coefficient of correlation, r.

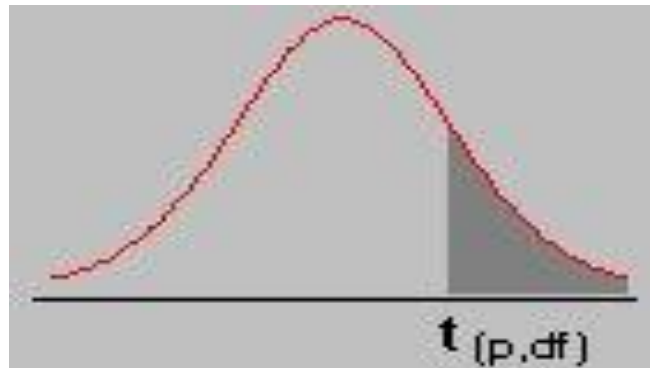
c) The coefficient of determination, r^2

[Appendix A: table of values for Standardized normal distribution \(z table\)](#)

0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4215	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4492	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857

2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998

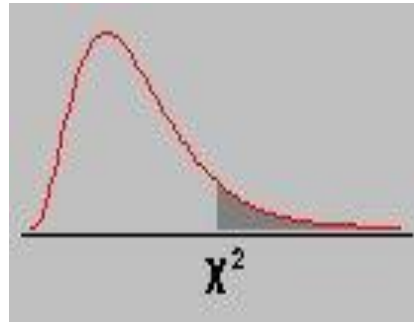
Appendix B: Table for t-Distribution with Right Tail Probabilities



t \ α =	0.1	0.05	0.025	0.01	0.005	0.0025	0.001	0.0005
df = 1	3.078	6.314	12.706	31.821	63.656	127.321	318.289	636.578
2	1.886	2.920	4.303	6.965	9.925	14.089	22.328	31.600
3	1.638	2.353	3.182	4.541	5.841	7.453	10.214	12.924
4	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	4.773	5.894	6.869
6	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.029	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587
11	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.437
12	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.318
13	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221
14	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.140
15	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073

t α=	0.1	0.05	0.025	0.01	0.005	0.0025	0.001	0.0005
Df=16	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015
17	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.965
18	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922
19	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.883
20	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.850
21	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.819
22	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.792
23	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.768
24	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.745
25	1.316	1.708	2.060	2.485	2.787	3.078	3.450	3.725
26	1.315	1.706	2.056	2.479	2.779	3.067	3.435	3.707
27	1.314	1.703	2.052	2.473	2.771	3.057	3.421	3.689
28	1.313	1.701	2.048	2.467	2.763	3.047	3.408	3.674
29	1.311	1.699	2.045	2.462	2.756	3.038	3.396	3.660
30	1.310	1.697	2.042	2.457	2.750	3.030	3.385	3.646
40	1.303	1.684	2.021	2.423	2.704	2.971	3.307	3.551
50	1.299	1.676	2.009	2.403	2.678	2.937	3.261	3.496
60	1.296	1.671	2.000	2.390	2.660	2.915	3.232	3.460
Infinity	1.282	1.645	1.960	2.326	2.576	2.807	3.090	3.290

Appendix C: Table for Right tail areas for the Chi-square Distribution



df\area	0.995	0.99	0.975	0.95	0.9	0.25	0.1	0.05	0.025	0.01	0.005
1	0.000	0.000	0.001	0.004	0.016	1.323	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	2.773	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	4.108	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	5.385	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	6.626	9.236	11.071	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	7.841	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	9.037	12.017	14.067	16.013	18.475	20.278
8	1.344	1.647	2.180	2.733	3.490	10.219	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	11.389	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	12.549	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	13.701	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	14.845	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	15.984	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	17.117	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	18.245	22.307	24.996	27.488	30.578	32.801

16	5.142	5.812	6.908	7.962	9.312	19.369	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	20.489	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	21.605	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	22.718	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	23.828	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	24.935	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.041	26.039	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	27.141	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	28.241	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	29.339	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	30.435	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	31.528	36.741	40.113	43.195	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	32.620	37.916	41.337	44.461	48.278	50.993
29	13.121	14.256	16.047	17.708	19.768	33.711	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.493	20.599	34.800	40.256	43.773	46.979	50.892	53.672